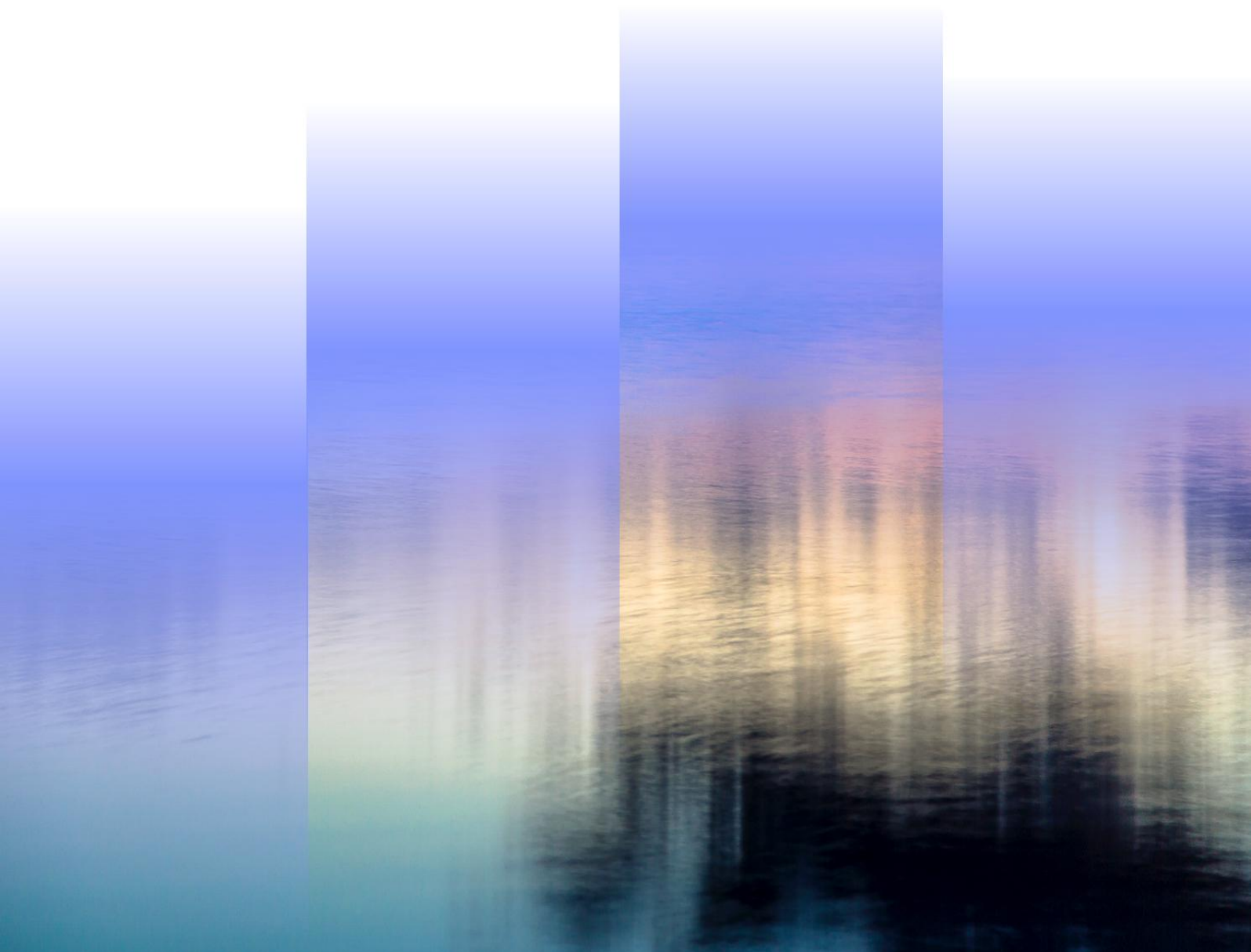


STATE OF THE ART OF WIRELESS CHARGING

INSIGHTS ON INDUSTRY TRENDS,
TECHNOLOGY STATUS, AND DIFFUSION

Mario Verdugo, Ville Naumanen
and Ville Tikka



EMRC Report: State of the art of wireless charging

Mario Verdugo, Ville Naumanen, Ville Tikka

LUT Scientific and Expertise Publications / Raportit ja selvitykset 143

ISSN-L 2243-3384, ISSN 2243-3384

Electronic publication: 978-952-412-405-8

Acknowledgements

The preparation of this report was funded by the Electric Mobility Research Center (EMRC) at LUT University. Special recognition is given to Ville Naumanen and Ville Tikka, co-authors of this report, for their valuable contributions, expertise, and strategic insights. Their input played a significant role in shaping the analysis, strengthening the technical depth, and enhancing the overall quality and relevance of this publication.

Contents

Executive summary	10
1. Introduction.....	13
1.1. How the technology works	13
1.2. Applications	14
Automation Potential in Mobility.....	14
Labor and Regulatory Considerations	15
Dynamic Wireless Charging	15
2. Wireless Charging Technology.....	16
2.1. System Architecture Comparison: Conductive and Wireless Charging	16
2.2. Wireless power transmission	17
2.3. Engineering aspects and challenges for wireless charging system design	17
Coil Coupling Efficiency and Misalignment Sensitivity.....	18
Wiring and Placement Constraints in High-Frequency Wireless Charging Systems	19
Environmental Protection and Packaging Requirements for Wireless Charging Components	19
Safety and Foreign Object Detection.....	20
State of the art of coil design	21
3. Standardization and Patent Maturity	23
3.1. Standardization Maturity.....	25
SAE J2954	25
IEC 61980	26
ISO 5474-4:2025	28
3.2. Patent Maturity.....	29
Patent Maturity by Organization	32
Patent Maturity by Technologies and Components.....	35
4. Market overview	39
4.1. Pilots.....	40
4.2. Market share distribution from technical perspective.....	41

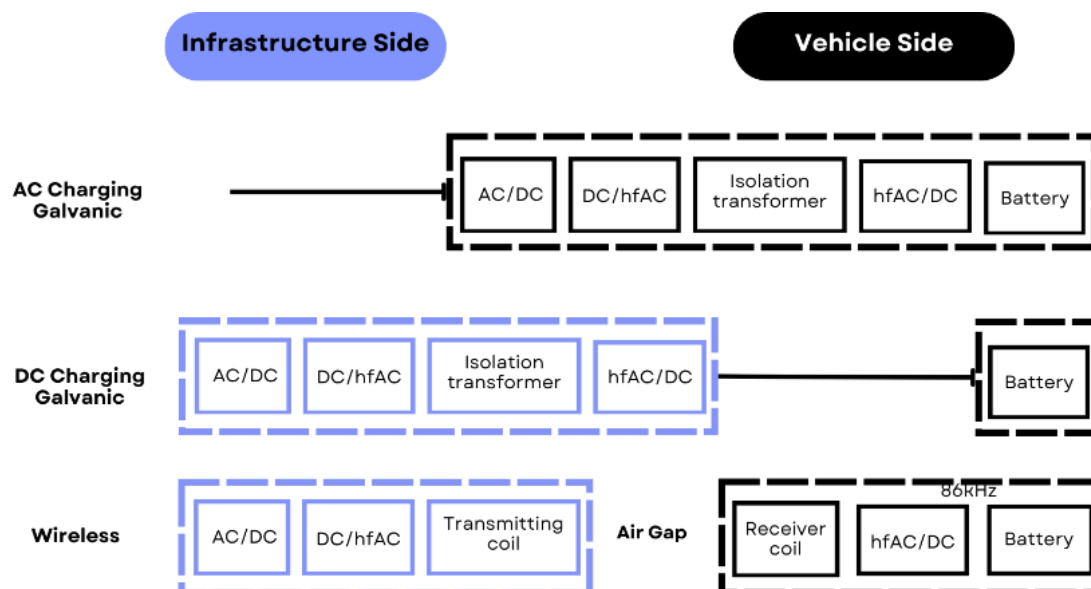
4.3.	Technology Maturity and Diffusion	41
4.4.	Market Players	43
	Geographical spread of the players	44
	Pure Players.....	45
5.	Summary.....	46
5.1.	Standardization and Regulatory Gaps	46
5.2.	Technical Challenges	46
5.3.	Market and Deployment Barriers	47
6.	Appendixes:.....	48

Executive summary

This report explores the current state of **wireless charging** for electric vehicles (EVs). Prepared in the format of a whitepaper, the report explains the working principles of wireless charging technologies. Additionally, the report assesses the technological and market development levels of this emerging trend. It analyzes the *market* and identifies *key players*, and examines the *diffusion levels* from *academic*, *commercial*, *standardization*, and *intellectual property perspectives* utilizing *public databases*.

Wireless charging transfers electrical energy from power grid to a vehicle without the need for physical connectors. The most promising technology, resonant inductive power transfer, bridges the gap between a *transmitting coil in the charging infrastructure* and a *receiver coil* integrated into the vehicle with high-frequency electromagnetic resonance.

Conventional conductive charging involves direct contact through cables and connectors, using AC or DC currents for power transfer. In contrast, wireless charging employs resonant inductive coupling, utilizing alternating magnetic fields to transfer energy without physical contact. These systems differ in energy efficiency, power density, integration, alignment, and safety protocols. Wireless charging air gap with transmitting and receiving coils are analogous to isolation transformer in galvanic charging applications.



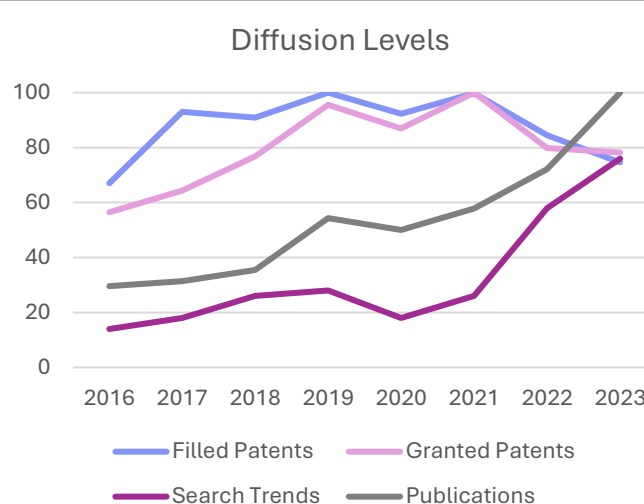
Main Differences: Wireless vs Conventional Charging

Feature	Conventional Charging	Wireless Charging
Energy Transfer Method	Galvanic contact via cable and connector	Contactless via magnetic induction or electric fields between capacitor plates
User interaction	Cable handling by user. Automation requires complex mechanics.	No need for user action, but vehicle positioning needs to be accurate for effective power transfer
Power Range	up to +1MW	3.7 kW – 22 kW (some pilots up to 270 kW)
Energy Efficiency	94-96%	44 -89% (Highly depended on positioning)
Standardization Maturity	Highly standardized and interoperable	First SAE standard published for low power. High power standards under development.

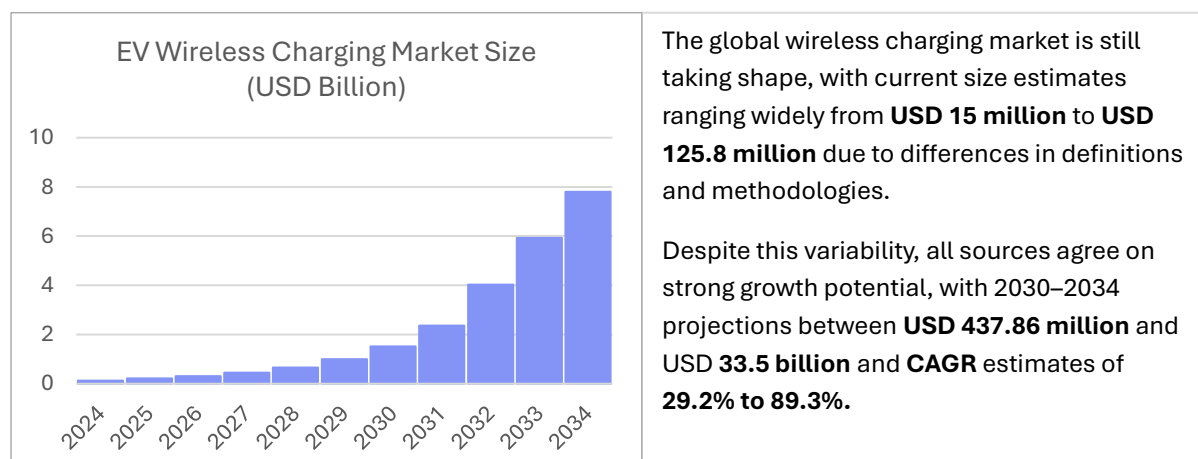
Maturity and Diffusion

While *public and academic interest* in wireless charging has **grown steadily**, driven by rising search trends and publications, patent activity has **slightly declined** since 2021, following a surge that began in 2010.

This suggests a gap between early technological development and slower market adoption. The drop in patenting may reflect industry hesitation due to market readiness, standardization challenges, cost, technical feasibility concerns, and safety issues.



Market Overview



Key Findings

- Patenting trend indicates a slight decline in industrial investments since 2021, while recent rising academic and search trends show growing public awareness towards wireless charging. Public awareness may indicate market opening.
- The Witricity business model, which focuses on IP licensing, has positioned the company as the leader in patents among Pure Players. Japanese and Korean OEMs, along with technology and IP suppliers, are at the forefront in the automotive industry. Additionally, German companies such as Mahle and Bosch are leading the way as OEM suppliers, representing an opportunity for European suppliers.
- Standards coverage is limited, which explains slow adoption of the technology. SAE J2954 defining low-power (<50kW) and static applications in passenger cars was published just recently. While international standards for high power applications are being defined with draft versions.
- Technologically system components such as coils, power electronics and control systems are widely patented, while dynamic charging and foreign object detection are still developing.
- Market size estimates an average market value of USD 130 million in 2024, with varying reports, but all agree on the sector's annual average growth of 52%.
- American tech start-up companies are leading market development with up to USD 14 million in funding for ongoing pilots and commercial products for transportation and industrial vehicles.
- Accuracy of Vehicle location on the charger has high impact on power throughput.
- Coil design, thermal management, electromagnetic compatibility and foreign object detection are key engineering challenges.

1. Introduction

Wireless charging technology is rapidly reshaping the way electric vehicles and automated machinery access power, presenting the prospect of seamless, contactless energy transfer and new opportunities for automation across sectors such as transportation, ports, and mining. By enabling charging without physical connectors, this innovation streamlines operations and enhances safety, while also laying the groundwork for improved efficiency and productivity as industries continue to advance toward autonomous workflows.

1.1. How the technology works

Wireless charging transfers electrical energy from a power source to a device without the need for physical connectors. With wireless charging EV charging can be initiated automatically without human interaction just by driving the vehicle close to the charger. Automatic charging is crucial for future autonomous vehicles but may also reduce the friction of EV adoption in heavy applications like in ports and mining.

The electromagnetic induction (recently resonant) bridges the gap between a transmitting *coil (pad)* in the charging infrastructure and a receiver *coil (unit)* integrated into the vehicle. Figure 1 showcases simplified concept of the wireless charging system. The main components perform the following process to achieve the charging:

1. **AC/DC Converter (Infrastructure)** – Converts alternating current (AC) from the grid into direct current (DC), preparing it for high-frequency conversion.
2. **DC/hfAC Converter (Infrastructure)** – Converts the DC into high-frequency alternating current (hfAC), which is more suitable for wireless energy transfer.
3. **Transmitting Coil (Infrastructure)** – Generates an alternating magnetic field that wirelessly transfers energy across the air gap to the vehicle's receiver coil.
4. **Receiver Coil (Vehicle)** – Captures the magnetic field from the transmitting coil and induces an alternating current within the vehicle.
5. **hfAC/DC Converter (Vehicle)** – Converts the high-frequency AC from the receiver coil back into DC current for storage.
6. **Battery (Vehicle)** – Stores the received DC electrical energy, making it available for powering the vehicle's motor and auxiliary systems.

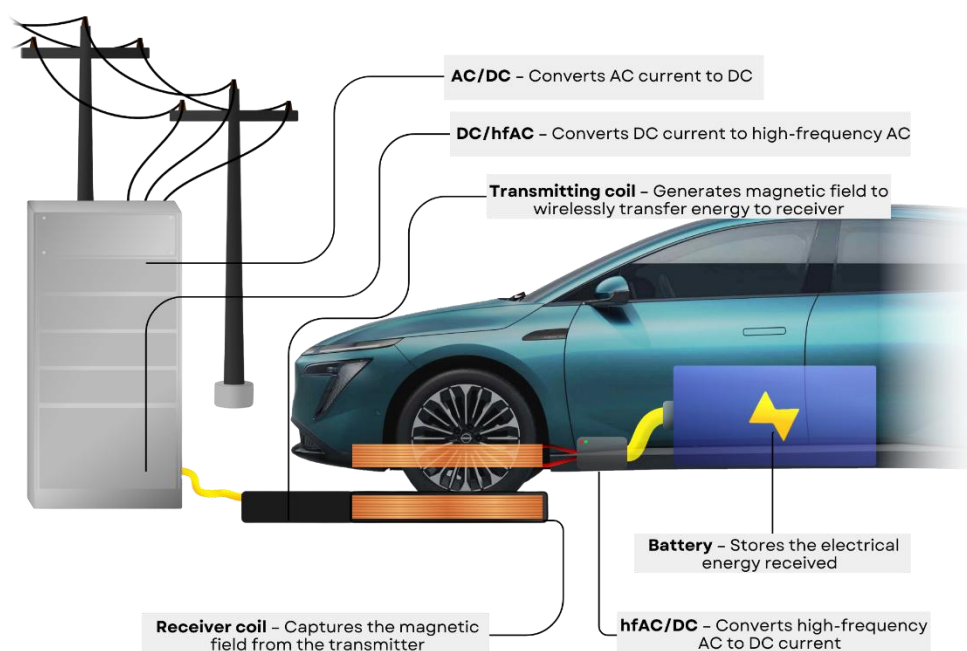


Figure 1. Simple concept illustration of a wireless charging system.

1.2. Applications

Wireless charging is increasingly viewed as a promising enabler of automation in mobility, primarily due to its ease of use and elimination of manual intervention. Unlike conductive systems that require physical connection via heavy-duty cables, wireless systems allow vehicles to charge simply by positioning over a transmitter pad—making them ideal for autonomous and semi-autonomous operations.

Automation Potential in Mobility

The ability to charge without human involvement opens the door to fully automated vehicle workflows, particularly in sectors where vehicles operate in closed environments.

Ports and mining sites are among the most advanced sectors in adopting automated machinery, including autonomous container handlers, haul trucks, and support vehicles. In these environments, vehicle uptime is a key performance metric. For example, container handlers may pause for several minutes while queuing or lifting containers.

Another example of suitable application is in automated guided vehicles (AGVs) and robotic material handlers which require seamless energy replenishment to maintain workflow continuity. Wireless charging enables incremental charging during short stops, such as loading, unloading, or waiting at intersections. With strategically placed wireless charging pads, these idle moments can be utilized for opportunity charging, allowing vehicles to harvest energy without interrupting operations. This not only improves fleet utilization but also reduces the need for oversized batteries and minimizes downtime associated with manual charging.

Labor and Regulatory Considerations

In some regions, labor union regulations restrict manual handling of high voltage charging cables, particularly for heavy trucks and industrial vehicles. These rules can create operational bottlenecks, require dedicated personnel or delays in charging schedules. Wireless charging offers a compliance-friendly alternative, bypassing manual connection and reducing labor dependency while improving safety.

This is especially relevant in sectors facing labor shortages or automation mandates, where minimizing human involvement in routine tasks like charging can yield significant productivity gains.

Dynamic Wireless Charging

While dynamic wireless charging, charging vehicles while in motion, presents a compelling vision for uninterrupted mobility, its widespread adoption faces significant hurdles:

High infrastructure investment: Embedding coils into roadways or transit corridors requires substantial capital and long-term planning.

Safety and societal acceptance: Concerns around electromagnetic exposure, system reliability, and road maintenance will slow public acceptance.

Standardization and interoperability: Dynamic systems require precise alignment, real-time communication, and robust standards, which are still evolving.

As a result, dynamic wireless charging remains in the early development phase, with pilot projects and limited commercial deployments primarily focused on low-power transit applications.

2. Wireless Charging Technology

Charging architectures differ primarily in the distribution of power conversion stages and the mechanism of energy transfer. Conventional conductive DC charging centralizes power conversion within the station, while wireless inductive charging distributes it between infrastructure and vehicle to enable contactless transfer via resonant magnetic coupling. This structural difference drives variations in efficiency, thermal behavior, EMC, and deployment strategies explored in the following sections.

2.1. System Architecture Comparison: Conductive and Wireless Charging

Basic system diagrams of traditional conductive DC charging and wireless inductive charging reveal a fundamental architectural difference. In a conventional DC charging system, the entire power conversion chain—from AC input to regulated DC output—is housed within the charging station infrastructure. This includes the AC-DC rectifier, DC-hfAC converter, an isolation transformer and hfAC-DC converter. The vehicle receives DC power directly via a cable and connector, with minimal onboard power electronics involved in the charging process.

In contrast, wireless charging systems distribute the power electronics between the infrastructure side and the vehicle side. The ground-based transmitter unit contains the AC-DC conversion, DC-hfAC converter and a transmitting coil, while the vehicle-mounted receiver includes a resonant coil, rectifier, and interface to the battery management system. This split architecture is necessary to facilitate inductive power transfer across an air gap, as it allows for efficient energy transmission without direct electrical contact between the two systems. Figure 2 illustrates the primary components of both systems mentioned above.

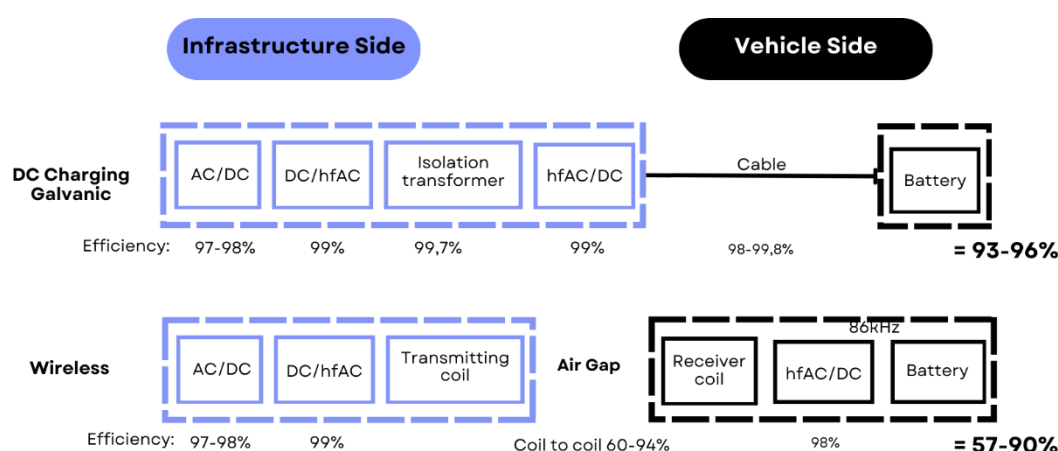


Figure 2. System architecture comparison of galvanic and wireless charging systems.

From a functional standpoint, the transmitter coil, receiver coil, and air gap in a wireless system can be conceptually compared to the charging cable and isolation transformer in a conductive system. Both serve to transfer energy while maintaining electrical isolation, but the wireless system does so via magnetic coupling rather than physical conductors.

2.2. Wireless power transmission

Three distinct wireless power transfer (WPT) technologies, inductive power transfer (IPT), resonant inductive power transfer (RIPT), and capacitive wireless power transfer (CWPT) are currently under exploration and development for electric vehicle applications.

Resonant Inductive Power Transfer (RIPT)

RIPT is the most widely adopted technology in commercial wireless charging systems today. It builds upon traditional inductive methods by using resonant circuits to enhance energy transfer over greater distances and misalignments. This makes RIPT more suitable for dynamic or semi-dynamic charging scenarios, such as charging while vehicles are stationary or moving slowly over embedded coils.

Inductive Power Transfer (IPT)

IPT represents the earlier generation of wireless charging technology. While it is simpler in design, it suffers from lower efficiency and requires precise coil alignment between the transmitter and receiver. These limitations have hindered its scalability and broader adoption, especially in fleet applications where operational flexibility is critical.

Capacitive Wireless Power Transfer (CWPT)

CWPT is an emerging technology that uses electric fields rather than magnetic fields to transfer energy. Although it offers potential advantages such as lightweight components and reduced electromagnetic interference, CWPT is still largely in the research phase. Its practical implementation faces challenges related to safety, power density, and environmental sensitivity, which currently limit its appeal to academic and experimental contexts.

2.3. Engineering aspects and challenges for wireless charging system design

Wireless charging introduces multiple engineering challenges that need to be addressed. Some major challenges in applying wireless charging technology are summarized in Figure 3. Wireless charging for electric vehicles presents a suite of engineering challenges, from maximizing coil-to-coil efficiency and managing misalignment sensitivity, to overcoming wiring, environmental, and safety constraints. As the technology evolves, designers must address thermal management, electromagnetic

compatibility, and the detection of foreign objects, all while ensuring reliable and efficient power transfer in diverse, real-world conditions.

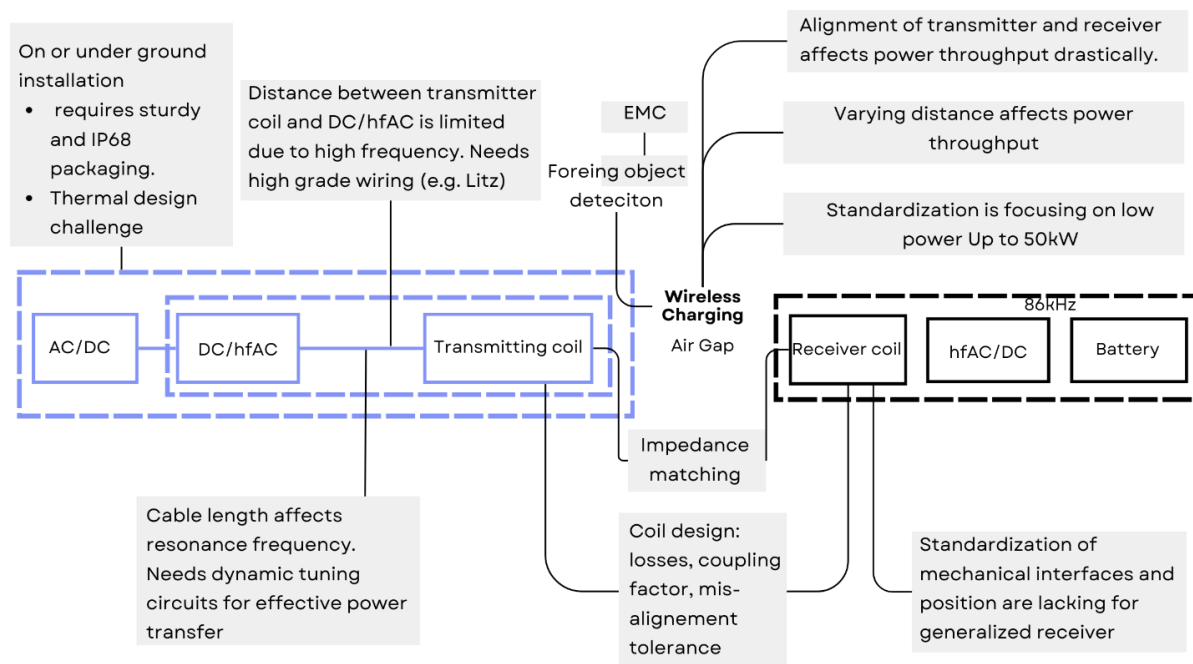


Figure 3. Summary of the main engineering challenges involved in the wireless charging applications.

Coil Coupling Efficiency and Misalignment Sensitivity

In wireless power transfer, the overall system efficiency and power transfer capability are predominantly governed by the coil-to-coil coupling efficiency. This efficiency is highly sensitive to:

- Lateral and angular misalignment between the transmitter and receiver coils.
- Air gap distance, which directly affects the magnetic coupling coefficient.

As the air gap increases, the mutual inductance between the coils decreases, leading to a significant drop in power transfer efficiency while simultaneously causing additional thermal stress to the components. For example, a well-aligned system with a 10 cm air gap may achieve >90% coil-to-coil efficiency, whereas a misaligned or extended gap can reduce efficiency below 60%, depending on the coil geometry and operating frequency.

This sensitivity necessitates precise alignment mechanisms and real-time control strategies to optimize performance. To mitigate the effects of misalignment and maintain efficient power transfer, compensation circuits are employed. These are passive circuits (typically using capacitors and inductors) designed to tune the system to its resonant frequency, minimize reactive power and maximize real power transfer and stabilize voltage and current waveforms under varying load and alignment conditions.

Wiring and Placement Constraints in High-Frequency Wireless Charging Systems

In wireless power transfer systems, particularly those operating at high frequencies (e.g., 85 kHz as specified by SAE J2954¹), the distance between the DC–hfAC converter and the transmitter coil must be minimized. This limitation arises from the frequency-dependent behavior of conductors and the electromagnetic characteristics of the transmission path. At high frequencies, phenomena such as skin effect, proximity effect, and parasitic inductance significantly increase losses and degrade signal integrity over longer distances.

These frequency-dependent effects challenge the stability and efficiency of the resonant inductive link. Longer wire lengths can shift the resonant frequency away from its optimal point, reducing power transfer efficiency. Impedance mismatches may arise, leading to increased reactive power and thermal losses. Also, electromagnetic emissions may increase, complicating EMC compliance. To compensate for these variations, resonant compensation circuit must be fine-tuned per installation. This tuning ensures that the system maintains resonance despite differences in wiring layout, component placement, or environmental factors. In installations where wire length cannot be precisely defined or standardized (e.g., modular or distributed systems), site-specific calibration becomes essential.

To mitigate these issues, high-grade conductors such as Litz wire are required but despite its advantages, it cannot fully compensate for the losses introduced by long transmission paths at high frequencies.

This constraint short cabling imposes limitations on system-level modularization, particularly in satellite-based charging architectures such as those used in Kempower-style distributed systems. In such designs, the DC–hfAC converter must be placed near the transmitter coil, effectively restricting the physical separation between power electronics and the coil to preserve efficiency and minimize thermal and electromagnetic losses.

Environmental Protection and Packaging Requirements for Wireless Charging Components

In wireless charging systems, particularly those designed for ground-level or underground installation, the DC–hfAC converter and transmitter coil assembly must be housed in mechanically sturdy IP68-rated enclosures. Because the transmitter coil is often embedded in or flush with the pavement, it needs to withstand standing water, dust and mechanical stress, including the risk of being overrun by vehicles.

¹ <https://sae.org>

Sealed and ruggedized enclosure sets constraints on thermal management. During high-power operation, DC–hfAC converter and transmitter coil generate substantial heat, which must be effectively dissipated to maintain system reliability and efficiency. This is particularly challenging in underground installations, where ambient temperatures may be elevated and airflow is restricted. In hot climates, the coil assembly is also exposed to solar radiation, which can significantly raise the surface temperature of the enclosure. This adds to the internal heat load and may push components beyond their safe operating limits.

Safety and Foreign Object Detection

Wireless charging systems for electric vehicles typically operate at 85 kHz, a frequency within the low-end of the radiofrequency (RF) spectrum. While this is classified as non-ionizing radiation—meaning it does not directly damage DNA or tissues like ionizing radiation (e.g., X-rays or gamma rays)—there are still important safety considerations, particularly regarding electromagnetic field (EMF) exposure and medical implants.

The biological effects of long-term exposure to 85 kHz magnetic fields are not yet fully understood, and research in this specific frequency range remains limited. This uncertainty has led to public concern and cautious regulatory oversight, especially as wireless charging systems become more powerful and widely deployed.

Wireless charging systems generate oscillating magnetic fields that can potentially interfere with implantable medical devices, such as pacemakers and insulin pumps. These devices may be sensitive to electromagnetic interference (EMI), which could affect their operation or safety.

Although the ICNIRP (International Commission on Non-Ionizing Radiation Protection) and the IEEE C95.1 standard have established guidelines for safe exposure to RF fields in the 100 kHz to 300 GHz range, it's important to note that wireless charging at 85 kHz falls just below this range. This gap contributes to uncertainty and highlights the need for further research.

To ensure safety in public and residential environments, wireless charging systems must be designed to contain magnetic fields within safe boundaries. This includes:

- Shielding and field shaping
- Foreign Object Detection (FOD)
- Charging interlocks and proximity sensors

Foreign Object Detection is a critical safety and performance feature in wireless power transfer systems. Since energy is transferred via magnetic fields across an air gap, any unintended object, especially metallic, placed between the transmitter and receiver coils can:

- Interfere with magnetic coupling,
- Cause localized heating, and
- Pose fire or safety hazards.

FOD systems must operate in real time, particularly during initial coil alignment and charging initiation. Integration with communication protocols such as SAE J2954 and ISO 15118 enables coordinated responses, including halting charging or alerting the user when a foreign object is detected. Technologies to detect foreign have not matured and further research is needed.

State of the art of coil design

Recent research has focused on the development of three-phase, two-layer coil architectures for wireless power transfer systems. These advanced coil designs enable improved magnetic field uniformity and coupling efficiency, particularly in high-power applications.

For instance, in a 100 kW wireless charging system, a coil-to-coil efficiency of 97.4% has been achieved. However, when considering the full system, including power electronics and conversion stages, the DC-to-DC efficiency drops to 90.8%, highlighting the impact of losses outside the coil interface.

A breakthrough was demonstrated by Oak Ridge National Laboratory (ORNL)² in June 2024, where researchers successfully achieved 270 kW wireless power transfer to a Porsche Taycan, setting a world record for electric light-duty vehicles. The system achieved an overall efficiency of 95%, with the following parameters:

- Air gap: 4.75 inches (120.65 mm)
- Coil diameter: 19 inches (482.6 mm)
- Misalignment tolerance: ± 4 inches (± 101.6 mm), equivalent to 21% in any direction

The system utilizes lightweight polyphase electromagnetic coupling coils, which generate a rotating magnetic field through phase windings, as illustrated in Figure 4. This design eliminates current ripple and field cancellation, enhancing power density and thermal performance and improves tolerance to misalignment. Yet high efficiency can be achieved in ideal operating conditions, the alignment and variation in airgap play significant role as Figure 5 illustrates. Efficiency peaks at ideal air gap and accurate alignment and decreases as air gap increases or alignment is off-center, also too narrow air gap may reduce coil-to-coil efficiency in case of the polyphase coil design.

² <https://www.ornl.gov/news/polyphase-wireless-power-transfer-system-achieves-270-kilowatt-charge-sets-another-world>

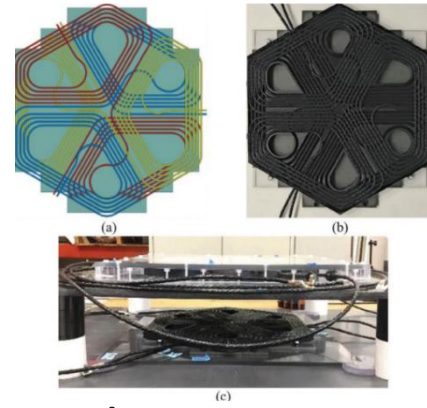
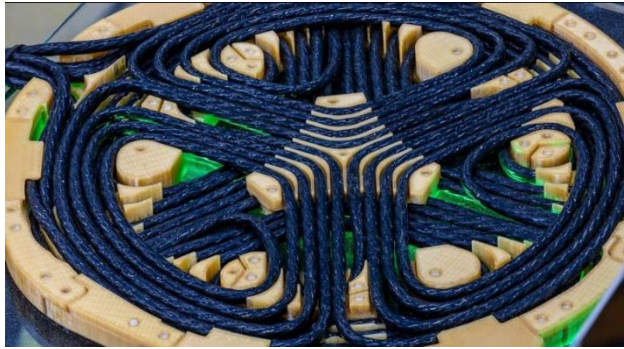


Figure 4. Two-layer, polyphase coil design³.

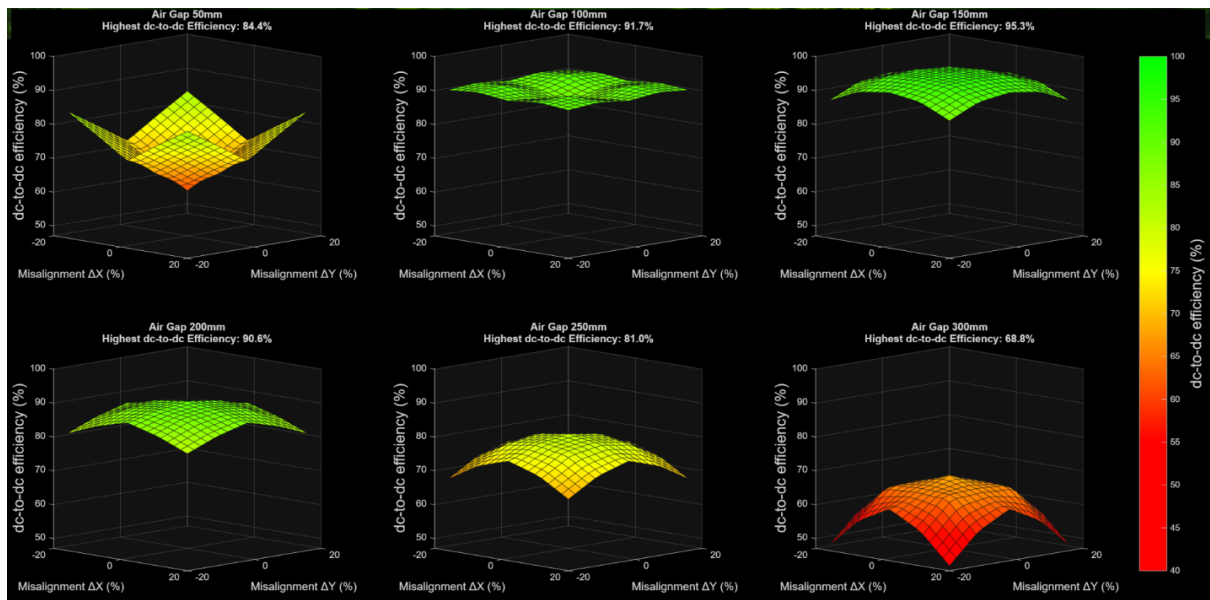


Figure 5. Efficiencies of a 50 kW polyphase coupler under misalignment at varying air gaps³.

³ Asa, E., Onar, O., Mohammad, M., Galigekere, V. P., Su, G. J., & Ozpineci, B. (2024). Overview of High-Power Wireless Charging Systems and Analysis of Polyphase Wireless Charging System Phase Winding and Resonant Tuning Network Connection Configurations. *IEEE Transactions on Transportation Electrification*.

3. Standardization and Patent Maturity

This report offers a holistic view of wireless charging landscape by outlining the technology but also fusing two critical lenses: patent activity and standard-development progress. Using the latest global patent data set and the current status of IEC⁴, ISO⁵, and SAE⁶ standards, it is analyzed the technology spaces are already crowded and the ones emerging. The analysis categorizes wireless charging technology into two perspectives: vehicle and infrastructure-side.

From the **patent perspective**, the analysis summarizes the present landscape of patent publications, including notable patent trends and existing gaps. It determines the maturity levels by the number of granted patents across various technologies and components, emphasizing both saturated and emerging areas. The patent maturity scale is divided into **saturated** ■ (> 50 patents), **mature** ■ (26-50 patents), **growth** ■ (11-25 patents), and **emerging** ■ (≤ 10 patents)

From the **standardization perspective**, the analysis outlines the key industry standards, the requirements for standardization, primary challenges, and immediate opportunities. It determines the maturity levels according to the stage of standard publication for each technology and its respective components. The standardization maturity scale can be divided into **highly published** ▲ (fully-published, widely cited), **medium published** ▲ (draft/partial specs), and **low published** ▲ (early WG stage / guidelines only).

Table 1 and maturity map (Appendix 1) visually illustrate the connection between patent density and standards progress across various technologies, providing an overview of mature, growing, and emerging technologies and their components, as well as the regulatory status within those fields.

⁴ <https://iec.ch/>

⁵ <https://www.iso.org/>

⁶ <https://www.iso.org/>

Table 1. Summary of standard maturity per technology area.

Domain	Segment	Patent maturity	Standards maturity
Vehicle	On-board Energy Capture	■ Saturated	▲ High (SAE J2954/2, IEC 61980-3)
	Alignment & Positioning	■ Saturated	▲ Medium (SAE tolerance classes)
	Vehicle Comms & Control	■ Saturated	▲ High
	Safety & FOD	■ Emerging	▲ Low (SAE draft Sec 9)
	Dynamic-Charging Pickup	■ Emerging	▲ Low (IEC PAS 61980-5)
Infrastructure	Ground-side Energy Transfer	■ Saturated	▲ High (SAE J2954/1, IEC 61980-2)
	Charging Station	■ Saturated	▲ High
	Grid Interface & Filters	■ Emerging	▲ Low (no dedicated clause)
	Alignment Aids (Ground)	■ Saturated	▲ Medium (no sensor-agnostic guide)
	Infra Safety & FOD	■ Emerging	▲ Low (SAE draft Sec 9)
	Dynamic In-road Segments	■ Emerging	▲ Low (IEC PAS 61980-5)

Although *Energy Transfer* technologies have matured in both the vehicle and infrastructure perspectives, thermal management and magnetic shielding are still in development, with potential for new developments.

Foreign Object Detection (FOD) and safety systems have been constantly developing and growing in the number of patents from both vehicle and infrastructure, including metallic detection and livable object detection, which are essential to ensure safety. Components include sensors, actuators, and control systems.

From the infrastructure side, alignment and guidance systems are still in their growing phase in terms of the number of patents and standard publishing status, representing a gap and opportunity for further developments.

Dynamic charging for both low and high power is the technology that has emerged in recent years, including its components for both the infrastructure and vehicle, such as embedded coils, sliding coils, and segment routers, among other components.

3.1. Standardization Maturity

For any charging technology to achieve widespread adoption, interoperability is essential. Standardization guarantees that any compatible vehicle can charge at any compliant infrastructure setting, regardless of the manufacturer. As a result, three standards have emerged that define the global market.

SAE J2954

Developed by SAE International, J2954⁷ is the definitive standard for North American and European markets. It establishes criteria for interoperability, safety, and performance for light-duty vehicles. The J2954 framework is expanding to address future market needs, with J2954/2 specifying power levels up to 500 kW for heavy-duty vehicles and J2954/3 addressing dynamic, in-motion charging. Table 2 illustrates the technical specifications of SAE J2954, including power levels, frequency, and alignment, among other specifications.

⁷ <https://www.sae.org/standards/content/j2954/>

Table 2. Technical specifications of SAE J2954, including power levels, frequency, and alignment, among other specifications.

Technology	SAE J2954
Status	SAE J2954 - Published Standard (Aug 2024) J2954/2 (Work in Process) - Technical Information Report (Dec 2022) J2954/3 (Work in Process) - Draft Development Phases
Power Levels	J2954: WPT1 (3.7 kW), WPT2 (7.7 kW), WPT3 (11 kW/kVA), WPT4 (22 kVA), WPT5 (50 kW) J2954/2: Up to WPT9 (500 kW)
Frequency	Nominal Band: 79–90 kHz Nominal Frequency: 85 kHz
Efficiency	Grid-to-Battery: ≥85% at center alignment; ≥80% within tolerance
Voltage Ratings	Up to 1000 V AC and 1500 V DC from the supply network.
Alignment Tolerance	Primary Device (GA): X-axis: ±75 mm, Y-axis: ±100 mm.
Communication and Control	Protocols: SAE J2847/6 (JSON-based), harmonizing with ISO 15118. Functions: Alignment, pairing, parameter negotiation, control, and termination of charging.
Dynamic Performance	Emergency Shutdown: Not explicitly detailed in snippets. Communication Loss: Managed via communication protocols (e.g., SAE J2847/6) to ensure safe shutdown.
Construction and Materials	Power Cable: Litz wire is used for coils to reduce AC losses. Insulation: Must meet dielectric strength and resistance requirements. Mechanical Strength: Must withstand vehicle rollover and environmental exposure.

IEC 61980

The International Electrotechnical Commission (IEC) is a global standard for all electrical and electronic technologies, the IEC provides a comprehensive and foundational framework for WPT systems. The IEC 61980⁸ series is structured as follows: (1) IEC 61980-1 covering general requirements, including electrical safety, applicable to all WPT

⁸ <https://iec.ch/>

technologies, (2) IEC 61980-2, which specifies the requirements for communication and system activities between the vehicle and the charging infrastructure, and (3) IEC 61980-3, which details the specific requirements for MF-WPT systems, including power transfer and EMC and IEC PAS 61980-5 defines interoperability and safety for dynamic charging. In addition, IEC PAS 61980-4 for high-power wireless power transfer standard draft is currently under development. IEC PAS 61980-4 is expected to proceed in final version preparation stage in autumn 2025. Table 3 summarizes the technical requirements of the IEC 61980 and ISO 5474.

Table 3. Technical specifications of IEC 61980 and ISO 5474, including power levels, frequency, and alignment, among other specifications.

Technology	IEC 61980 and ISO 5474
Status	IEC 61980 - Published Standards (2020-2023) PAS for high-power and dynamic - Draft Phase
Power Levels	IEC 61980-3: WPT1 (≤ 3.7 kW), WPT2 (≤ 7.7 kW), WPT3 (≤ 11.1 kW). IEC PAS 61980-4: MF-WPT4 (11.1–22 kW), MF-WPT5 (22–50 kW), MF-WPT6 to MF-WPT9 (up to 500 kW)
Frequency	Nominal Band: 79-90 kHz Nominal Frequency: 85 kHz Stability: ± 0.050 kHz during a charging session
Efficiency	Grid-to-Battery: $\geq 85\%$ at center alignment; $\geq 80\%$ within tolerance for WPT1-3.
Voltage Ratings	Up to 1000 V AC and 1500 V DC from the supply network.
Alignment Tolerance	Supply Device: X-axis: ± 75 mm, Y-axis: ± 100 mm. EV Device: Must operate within the tolerance defined by IEC 61980-3 (X: ± 75 mm, Y: ± 100 mm).
Communication and Control	Protocols: IEC 61980-2, specifies communication based on ISO 15118. Functions: Start, control, and terminate WPT process.
Dynamic Performance	Emergency Shutdown: Within 1 second. Communication Loss: Power reduction within 4 seconds, followed by full shutdown to 0 W.
Construction and Materials	Power Cable: Litz wire recommended. Insulation: Resistance and dielectric strength must be verified per IEC 61980-1. Mechanical Strength: As per IEC 61980-1 Clause 14.

ISO 5474-4:2025

The International Electrotechnical Commission (IEC) and the International Organization for Standardization (ISO) have adopted a modular and collaborative framework for WPT standardization. By mutual agreement between their respective technical committees (IEC/TC 69 and ISO/TC 22), the system requirements have been merged, creating a future

joint ISO/IEC standard. The IEC 61980 series of standards govern the off-board infrastructure (the supply device or charging station), while the ISO 5474-4:2025⁹ standard governs the on-board vehicle equipment. This division of labor is the defining philosophical characteristic of the IEC/ISO approach, creating a distinct ecosystem with its own set of advantages and challenges.

3.2. Patent Maturity

This section examines the patent maturity landscape wireless charging technologies through a patent analysis from patent databases^{10 11}, highlighting their development stages, innovation trends, and emerging gaps. There has been a considerable decline in patents related to wireless charging. From 2007 to 2021, there was a constant growth in patent applications from different perspectives, such as coil development, static charging methods, and alignment systems, as illustrated in Figure 9. Despite the decline, there are still aspects that experience a considerable growth, such as dynamic charging, high-power applications, Foreign Object Detection (FOD) systems, and adaptive power supply systems, among other innovations.

As patent applications decline, so do the granted patents. The patent applicants have been consolidated into a smaller group of technology leaders. However, as high-power and dynamic charging adoption increases in industrial applications, it leaves a gap for new patent applications.

The United States is the leader in granted patents in wireless charging technologies. Japan has been the technology ground-breaker along with Germany and other European countries. However, China and South Korea have been the countries with the most patents in recent times, particularly within the last decade.

⁹ <https://www.iso.org/standard/81300.html>

¹⁰ <https://worldwide.espacenet.com/>

¹¹ <https://patents.google.com/>

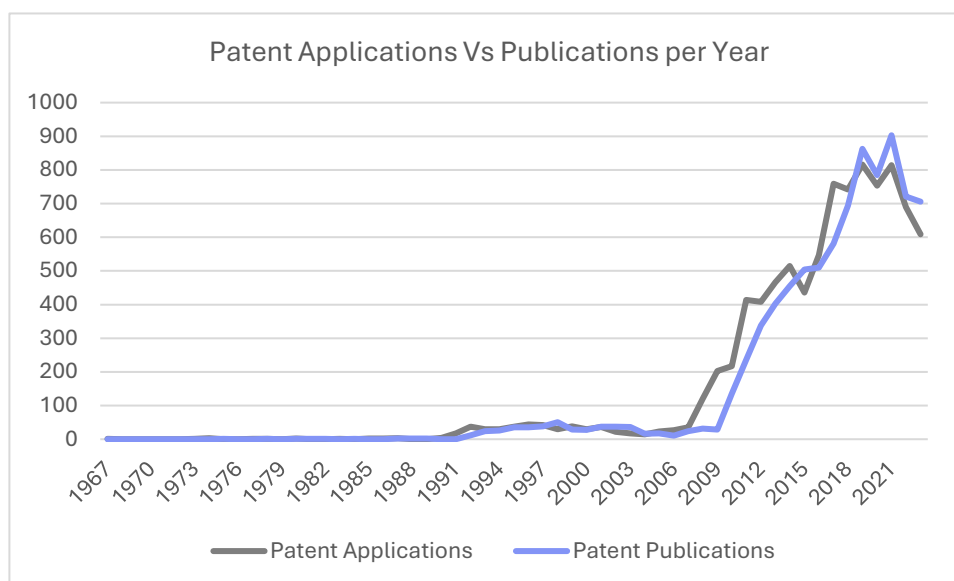


Figure 9. Wireless charging-related yearly patent applications and publications

The history of wireless charging patents encompasses a span of five decades, reflecting an evolution from rudimentary static systems to sophisticated high-power dynamic solutions. The inaugural patent for wireless power transfer for electric vehicles (EVs) was filed in 1975, with Japanese automakers, including Toyota, Honda, and Mitsubishi, emerging as pioneers in this field during the 1980s. The 1990s witnessed advancements in dynamic charging systems, the implementation of cooling technologies, and the development of self-alignment mechanisms. In the 2000s, the emphasis shifted toward improvements in coil design, signaling processes, and adaptive control systems. The 2010s introduced significant innovations such as foreign object detection and the integration of renewable energy sources. In 2010, Qualcomm Halo, which was later acquired by WiTricity, made a breakthrough in the technology by obtaining a patent related to magnetic inductive resonance. Since 2020, there has been a marked increase in high-power dynamic applications across various modes of transportation, highlighting the rapid advancements within this sector. Figure 10 illustrates the progression of this technology throughout its patent history.

First EV wireless power transfer patent

1975



Magnetic coupling device for power transmission (**Honda**) and ferromagnetic shell core for electric coils (**Porsche**)

Japanese companies take patent lead

1985 - 1990



Noncontact power supply for moving object (**Daifuku**) and noncontact current collector of running vehicle (**Sumitomo**)

Patent applicants expand worldwide

1995



Systems for transmitting energy and signals (**Bosch**), automakers patent coil units (**BMW, Nissan, Honda**)

Focus on signaling and adaptive control

2005 - 2010



Qualcomm (WiTricity today) first patented its Wireless Energy Transfer Using Repeater Resonators Patent

FOD and renewable energy intergation

2010 - 2015

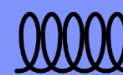
Wireless charger for underwater vehicles (Utah University), wireless charging for industrial vehicles (KAIST)



Focus on dynamic and high-power applications and integration

2020 - 2025

Inductive energized cables laid in the road to power an electrically driven vehicle.



First patents filled by automakers

1985

First patents on wireless chargers for EVs from **Toyota, Hitachi, Honda, Mitsubishi, and Yamaha.**



First patents on dynamic charging

1990 - 1995

First patents on cooling (**Hughes Aircraft**) and self-aligning systems (**Delco Electronics**)



Focus on transmission and receiver coils

2000 - 2005

Power control unit (University of Auckland), power supply system (KAIST) and power control device (Panasonic)



Magnetic Inductive Resonance Breakthrough

2010



Wind/solar wireless power transmission (Tianjin Polytechnic), system to detect foreign objects (Bombardier)

Expansion in aerial, maritime and industrial vehicles

2015 - 2020



Dynamic transmitting track (BYD), high-power dynamic wireless charging (Shenzhen Polytechnic).

Figure 10. Wireless Charging Patent Timeline

Patent Maturity by Organization

The patent landscape in wireless charging reveals distinct innovation patterns among different organizational roles. Diverse players with distinct roles and strategic focus shape the market for EV wireless charging. These can be categorized into four main groups:

- **Pure Players**, which are startups or SMEs, are fully dedicated to developing and commercializing wireless charging solutions.
- **Traditional OEMs** are established automakers where wireless charging remains a peripheral or experimental activity.
- **Technology and Component Suppliers**, OEMs supplying companies that offer wireless charging modules as part of a broader portfolio of automotive systems.
- **Infrastructure and IP Providers**, firms that supply critical intellectual property, embedded systems, or large-scale infrastructure to enable wireless charging deployments.

Pure Players focus primarily on wireless charging, driving rapid advancements. Traditional OEMs balance wireless charging research with broader electrification strategies. Technology and Component Suppliers enable hardware and system integration, while Infrastructure and IP Providers, often research institutions, expand fundamental technology and support large-scale implementation.

In the **Pure Players** category, companies focus almost entirely on wireless EV charging. WiTricity (Qualcomm) leads in resonance technology, specializing in FOD, sensors, circuits, and coil alignment. Beijing Invispower works on FOD, multi-frequency charging, and autonomous alignment; Anjie Wireless on high-power modular systems with advanced control and cooling; WiPowerOne on power control, dynamic charging, and flexible coils; and InductEV on resonant induction, FOD, harmonic signaling, and multi-vehicle charging. Table 4 details their patent activity and focus.

Traditional original equipment manufacturers (OEMs), such as Toyota, Honda, Nissan, BMW, Mercedes-Benz, Hyundai, and Ford are integrating wireless charging into their electrification and mobility strategies. Japanese OEMs have been active since the mid-1980s, establishing strong patents and focusing on dynamic charging and autonomous software for electric vehicle (EV) ecosystems.

Table 4. Pure Players patent behavior, activity and focus.

Segment	Patent Trend	Patents - Lastest Developments																														
Witricity Qualcomm (US)	<table><thead><tr><th>Year</th><th>Patents</th></tr></thead><tbody><tr><td>2012</td><td>20</td></tr><tr><td>2013</td><td>13</td></tr><tr><td>2014</td><td>27</td></tr><tr><td>2015</td><td>31</td></tr><tr><td>2016</td><td>31</td></tr><tr><td>2017</td><td>32</td></tr><tr><td>2018</td><td>35</td></tr><tr><td>2019</td><td>25</td></tr><tr><td>2020</td><td>8</td></tr><tr><td>2021</td><td>3</td></tr><tr><td>2022</td><td>2</td></tr><tr><td>2023</td><td>5</td></tr><tr><td>2024</td><td>4</td></tr><tr><td>2025</td><td>0</td></tr></tbody></table>	Year	Patents	2012	20	2013	13	2014	27	2015	31	2016	31	2017	32	2018	35	2019	25	2020	8	2021	3	2022	2	2023	5	2024	4	2025	0	Patents: 228 Last patents: FOD, sensing, and coil pairing.
Year	Patents																															
2012	20																															
2013	13																															
2014	27																															
2015	31																															
2016	31																															
2017	32																															
2018	35																															
2019	25																															
2020	8																															
2021	3																															
2022	2																															
2023	5																															
2024	4																															
2025	0																															
Beijing Invispower (CN)	<table><thead><tr><th>Year</th><th>Patents</th></tr></thead><tbody><tr><td>2012</td><td>0</td></tr><tr><td>2013</td><td>0</td></tr><tr><td>2014</td><td>0</td></tr><tr><td>2015</td><td>0</td></tr><tr><td>2016</td><td>0</td></tr><tr><td>2017</td><td>0</td></tr><tr><td>2018</td><td>1</td></tr><tr><td>2019</td><td>15</td></tr><tr><td>2020</td><td>25</td></tr><tr><td>2021</td><td>18</td></tr><tr><td>2022</td><td>3</td></tr><tr><td>2023</td><td>1</td></tr><tr><td>2024</td><td>0</td></tr><tr><td>2025</td><td>0</td></tr></tbody></table>	Year	Patents	2012	0	2013	0	2014	0	2015	0	2016	0	2017	0	2018	1	2019	15	2020	25	2021	18	2022	3	2023	1	2024	0	2025	0	Patents: 54 Last patents: alignment, and frequency adjustment.
Year	Patents																															
2012	0																															
2013	0																															
2014	0																															
2015	0																															
2016	0																															
2017	0																															
2018	1																															
2019	15																															
2020	25																															
2021	18																															
2022	3																															
2023	1																															
2024	0																															
2025	0																															
Anjie Wireless Tehcnology (CN)	<table><thead><tr><th>Year</th><th>Patents</th></tr></thead><tbody><tr><td>2012</td><td>0</td></tr><tr><td>2013</td><td>0</td></tr><tr><td>2014</td><td>0</td></tr><tr><td>2015</td><td>0</td></tr><tr><td>2016</td><td>0</td></tr><tr><td>2017</td><td>0</td></tr><tr><td>2018</td><td>0</td></tr><tr><td>2019</td><td>15</td></tr><tr><td>2020</td><td>25</td></tr><tr><td>2021</td><td>5</td></tr><tr><td>2022</td><td>6</td></tr><tr><td>2023</td><td>2</td></tr><tr><td>2024</td><td>1</td></tr><tr><td>2025</td><td>1</td></tr></tbody></table>	Year	Patents	2012	0	2013	0	2014	0	2015	0	2016	0	2017	0	2018	0	2019	15	2020	25	2021	5	2022	6	2023	2	2024	1	2025	1	Patents: 49 Last patents: cooling materials and systems, and high-power charging.
Year	Patents																															
2012	0																															
2013	0																															
2014	0																															
2015	0																															
2016	0																															
2017	0																															
2018	0																															
2019	15																															
2020	25																															
2021	5																															
2022	6																															
2023	2																															
2024	1																															
2025	1																															
WiPowerOne (KR)	<table><thead><tr><th>Year</th><th>Patents</th></tr></thead><tbody><tr><td>2012</td><td>0</td></tr><tr><td>2013</td><td>0</td></tr><tr><td>2014</td><td>0</td></tr><tr><td>2015</td><td>0</td></tr><tr><td>2016</td><td>0</td></tr><tr><td>2017</td><td>0</td></tr><tr><td>2018</td><td>0</td></tr><tr><td>2019</td><td>0</td></tr><tr><td>2020</td><td>3</td></tr><tr><td>2021</td><td>2</td></tr><tr><td>2022</td><td>10</td></tr><tr><td>2023</td><td>7</td></tr><tr><td>2024</td><td>3</td></tr><tr><td>2025</td><td>2</td></tr></tbody></table>	Year	Patents	2012	0	2013	0	2014	0	2015	0	2016	0	2017	0	2018	0	2019	0	2020	3	2021	2	2022	10	2023	7	2024	3	2025	2	Patents: 23 Last patents: dynamic charging, and multiple supply coils for industry.
Year	Patents																															
2012	0																															
2013	0																															
2014	0																															
2015	0																															
2016	0																															
2017	0																															
2018	0																															
2019	0																															
2020	3																															
2021	2																															
2022	10																															
2023	7																															
2024	3																															
2025	2																															
InductEV (US)	<table><thead><tr><th>Year</th><th>Patents</th></tr></thead><tbody><tr><td>2012</td><td>0</td></tr><tr><td>2013</td><td>0</td></tr><tr><td>2014</td><td>0</td></tr><tr><td>2015</td><td>0</td></tr><tr><td>2016</td><td>0</td></tr><tr><td>2017</td><td>0</td></tr><tr><td>2018</td><td>0</td></tr><tr><td>2019</td><td>0</td></tr><tr><td>2020</td><td>0</td></tr><tr><td>2021</td><td>3</td></tr><tr><td>2022</td><td>4</td></tr><tr><td>2023</td><td>4</td></tr><tr><td>2024</td><td>1</td></tr><tr><td>2025</td><td>4</td></tr></tbody></table>	Year	Patents	2012	0	2013	0	2014	0	2015	0	2016	0	2017	0	2018	0	2019	0	2020	0	2021	3	2022	4	2023	4	2024	1	2025	4	Patents: 16 Last patents: FOD, alignment and harmonic signaling.
Year	Patents																															
2012	0																															
2013	0																															
2014	0																															
2015	0																															
2016	0																															
2017	0																															
2018	0																															
2019	0																															
2020	0																															
2021	3																															
2022	4																															
2023	4																															
2024	1																															
2025	4																															

German manufacturers BMW and Mercedes-Benz were also early adopters. However, they have reduced their patenting activity decline since 2019. In contrast, Hyundai has made steady progress in dynamic charging and thermal management since 2015, while Ford contributed to bidirectional charging until 2020. Table 5 illustrates the patent behavior, activity and focus of the main OEMs. Table 5 shows the patent activity and focus of OEMs.

Table 5. OEMs patent behavior, activity and focus.

Segment	Patent Trend	Patents - Latest Developments
Toyota Motor (JP)	Line graph showing Toyota Motor (JP) patent activity from 2012 to 2024. The y-axis ranges from 0 to 100. The graph shows a general upward trend with a significant peak in 2023.	Patents: 793 Last patents: resonant transfer, dynamic charging, power control.
Hyundai Motor (KR)	Line graph showing Hyundai Motor (KR) patent activity from 2012 to 2024. The y-axis ranges from 0 to 30. The graph shows a fluctuating trend with a peak in 2023.	Patents: 213 Last patents: FOD, microscopic alignment, dynamic charging.
Honda Motor (JP)	Line graph showing Honda Motor (JP) patent activity from 2012 to 2025. The y-axis ranges from 0 to 30. The graph shows a fluctuating trend with a peak in 2021.	Patents: 117 Last patents: alignment, FOD, and coupling improvement.
Nissan Motor (JP)	Line graph showing Nissan Motor (JP) patent activity from 2012 to 2025. The y-axis ranges from 0 to 40. The graph shows a sharp peak in 2014 followed by a decline.	Patents: 113 Last patents: FOD, positioning guidance, and power control.
BMW (DE)	Line graph showing BMW (DE) patent activity from 2012 to 2025. The y-axis ranges from 0 to 30. The graph shows a peak in 2019 followed by a decline.	Patents: 96 Last patents: FOD, positioning guidance, and power control.
Mercedes Benz (DE)	Line graph showing Mercedes Benz (DE) patent activity from 2012 to 2025. The y-axis ranges from 0 to 40. The graph shows a peak in 2019 followed by a decline.	Patents: 96 Last patents: thermal energy management and coupling improvement.
Ford Global (US)	Line graph showing Ford Global (US) patent activity from 2012 to 2025. The y-axis ranges from 0 to 20. The graph shows a peak in 2018 followed by a decline.	Patents: 92 Last patents: self-locating charging and coupling improvement.

Technology and Component Suppliers play a critical role in enabling wireless charging by developing the hardware, control systems, and integration technologies that make large-scale deployment possible. Japanese leaders such as Denso and IHI have remained active in recent years, focusing on abnormality diagnosis systems, dynamic charging power control, and robust system integration for varying vehicle platforms. Sumitomo Electric, although less active since 2017, has contributed foundational work in conductive materials and coil systems. European suppliers, including Mahle, Bosch, Sew Eurodrive, and Siemens, have historically been IP leaders, particularly before 2020, advancing wireless charging stations, high-efficiency converters, and automation solutions for plugless charging processes. Their more recent developments emphasize station-to-vehicle communication protocols, modular pad designs, and scalable installation concepts for urban and fleet applications. This group's contributions are essential for bridging the gap between concept and commercial readiness, ensuring interoperability, safety, and manufacturability across diverse automotive markets.

Infrastructure and IP Providers are dominated by South Korean and Chinese institutions, with Korea's Korea Advanced Institute of Science and Technology, LG Group, and Samsung Electronics driving movable pad systems, multi-coil designs, and dynamic applications, while Chinese universities like Harbin Institute of Technology and the University of Chongqing lead in coupling mechanisms, bidirectional charging, and grid-integration technologies.

Patent Maturity by Technologies and Components

The patent maturity by the core technologies presented above presents the trends and current gaps in their main components, such as energy transfer, alignment systems, dynamic charging, communications, and safety systems through historical patent data.

Energy Capture and Transfer

The energy capture and transfer represent the essential technologies for wireless charging, with the receiver coil (vehicle) and transmitting coil (infrastructure) being the most relevant components. These technologies and most of their components are currently at mature status. However, components and subcomponents related to magnetic shielding and thermal management are still in a growing phase, leaving a gap for future patents. Below, the patent behavior of the energy capture and transfer over time is illustrated. Figures 11 to 15 show the patent evolution of energy capture and transfer components through years.

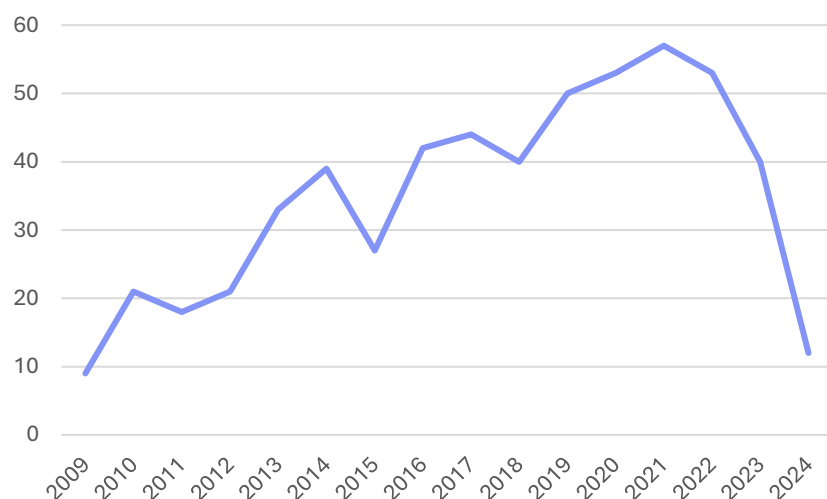


Figure 11. Receiver and Transmitting Coil Patent Applications

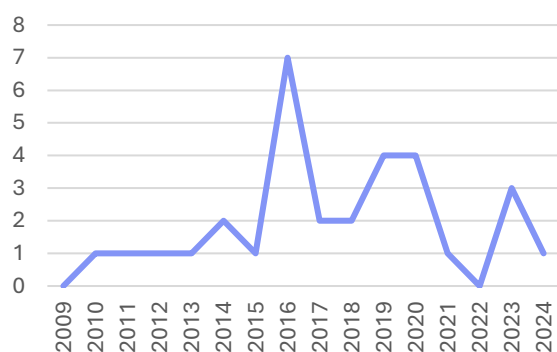


Figure 12. On-board rectifier Patent Applications

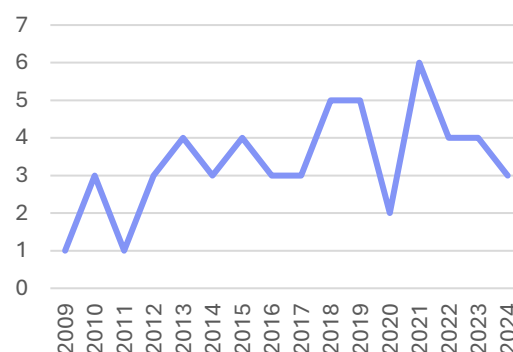


Figure 13. Converter/inverter Patent Applications

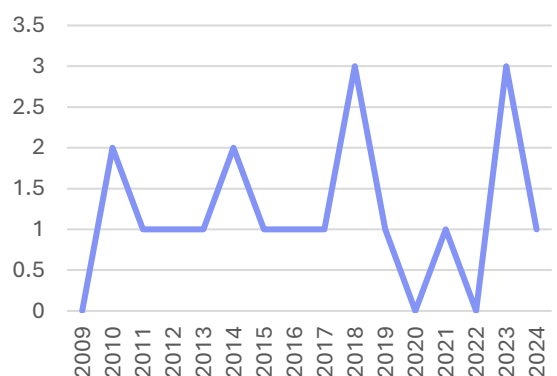


Figure 14. Thermal Management Patent Applications

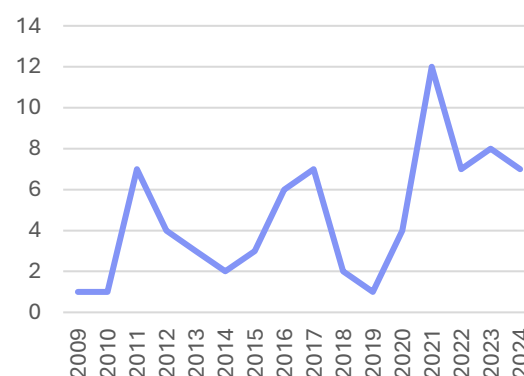


Figure 15. Power Electronics Patent Applications

Alignment Systems

Alignment, positioning, and guidance systems have emerged since the previous decade. However, technologies and components related to achieving an optimal alignment, and thus efficiency, are still of interest to organizations due to the ongoing patent applications. Figure 16 illustrates the patent evolution of alignment systems components through years.

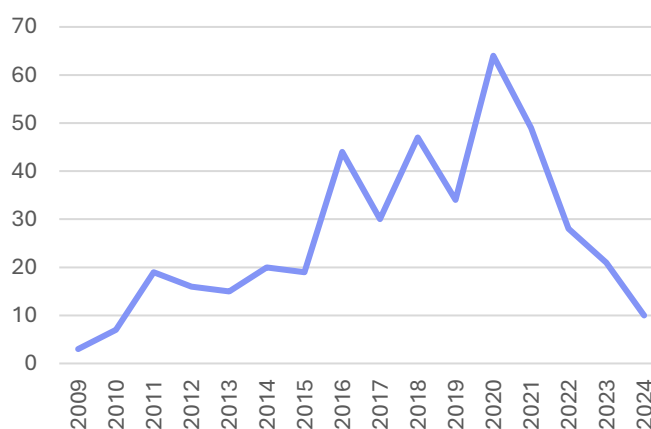


Figure 16. Alignment Systems Patent Applications

Dynamic Charging

Despite having several technical and adoption challenges, dynamic charging (and related components) is considered the most emerging technology in wireless charging. Patents in this technology have been granted in recent years as new applications in high-power and industrial settings grow. Figures 17 illustrates the patent evolution of dynamic charging components through years.

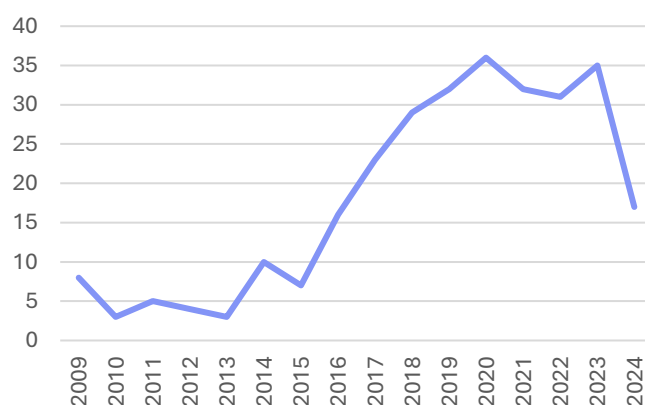


Figure 17. Dynamic Charging Patent Applications

Communications and Control

Communications and control systems are considered among the most mature technologies in terms of patents and standards. However, as new applications for both dynamic and high-power continue growing, patents for new systems that communicate and control the power are yet to be filed. Figure 18 illustrates the patent evolution of communication and control components through years.

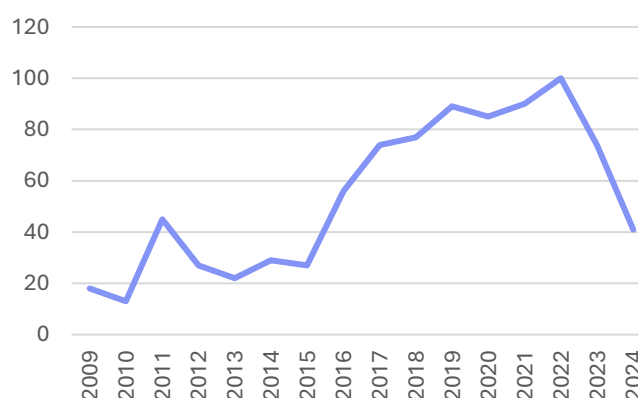


Figure 18. Communications and Control Patent Applications

Foreign Object Detection (FOD) and Safety Systems

One of the main concerns in achieving the successful implementation of wireless charging applications lies in the safety and risk issues. The safety issues are related to the transmission of high power through the wireless charging air gap that can cause damage to people and animals. On the risk side, a metallic object near the charging can significantly affect the efficiency and effectiveness of charging. Figures 19 illustrates the patent evolution of FOD and safety systems components through years.

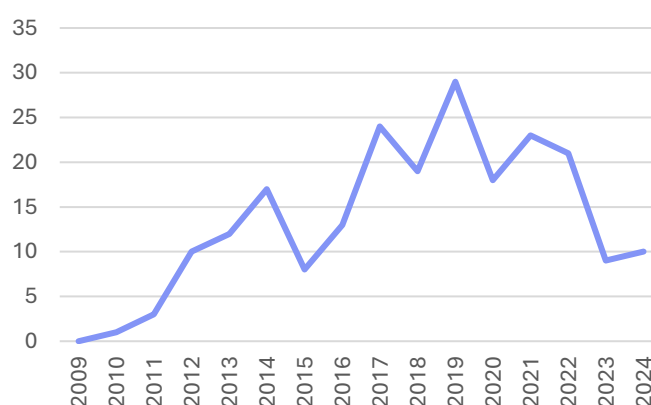


Figure 19. FOD and Safety Systems Patent Applications

4. Market overview

The global market for wireless charging in electric mobility is still in its early stages of definition, as evidenced by the wide variability in market size estimates reported by different research institutes and consulting firms¹². This inconsistency reflects differences in methodologies, scope of analysis, and definitions but is also a clear indication of non-matured technology of what constitutes the wireless charging market.

Reported latest market sizes range from **USD 15 million** to **USD 125.8 million**, highlighting the lack of consensus on the current scale of the industry. These discrepancies are largely due to varying assumptions about technology readiness, geographic coverage, and inclusion of pilot projects or early-stage deployments.

Despite the differences in current valuations, all sources agree on the significant growth potential of wireless charging technologies. The market is expected to expand rapidly as standards mature, infrastructure investments increase, and OEMs begin integrating wireless systems into mainstream vehicle platforms. Forecasts for 2030 to 2034 vary widely, with projected market values ranging from **USD 437.86 million** to **USD 33.5 billion**. The compound annual growth rate (CAGR) is estimated to be between **29.2%** and **89.3%**, depending on the scenario and market segment considered.

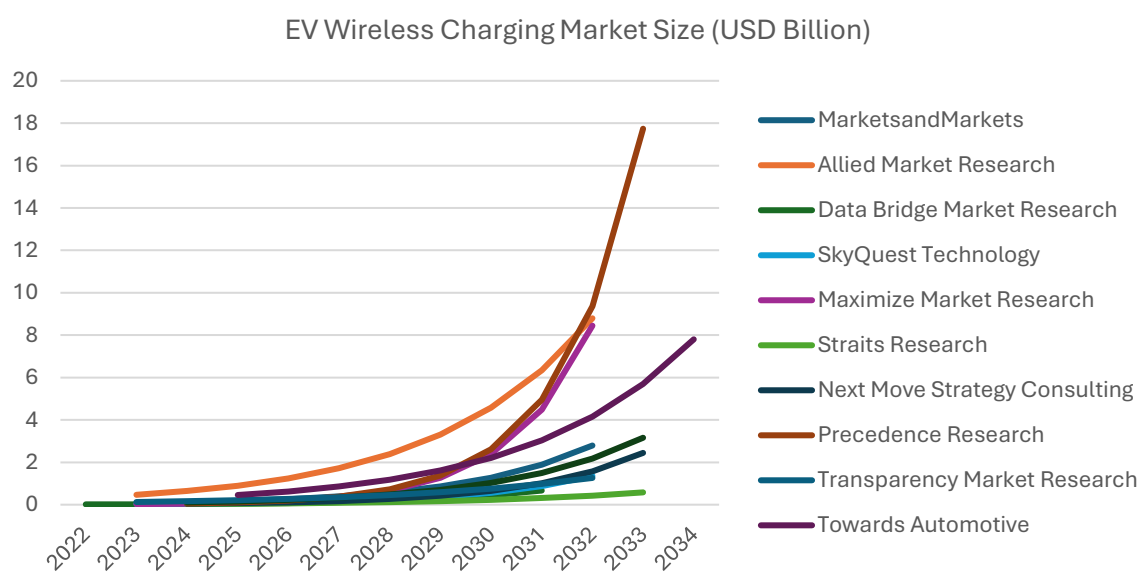


Figure 20. EV Wireless Charging market size and growth by research and consulting firms.

¹² <https://www.alliedmarketresearch.com/>; <https://www.databridgemarketresearch.com/>;
<https://www.marketsandmarkets.com/>; <https://www.maximizemarketresearch.com/>;
<https://www.nextmsc.com/>; <https://www.precedenceresearch.com/>; <https://www.skyquestt.com/>;
<https://straitsresearch.com/>; <https://www.towardsautomotive.com/>;
<https://www.transparencymarketresearch.com/>

Technology has the potential to simplify charging workflows, reduce labor dependency, and support dynamic energy management positions it as a strategic component in the future of transportation.

4.1. Pilots

Another notable trend in the evolution of wireless charging technology is its increasing deployment at higher power levels and across larger fleets. What began as small-scale demonstrations has now expanded into ambitious pilot programs involving dozens of electric buses, showcasing the technology's readiness for real-world transit operations. Appendices 2 to 6¹³ illustrate ongoing and deployed wireless charging pilot projects conducted by **Pure Players** that have successfully achieved significant milestones in the field. However, the following example shows additional pilot projects implemented for several applications in different countries.

UCLA's BruinBus Fleet (California, USA)

UCLA is implementing California's first in-road wireless EV charging system in partnership with Electreon¹⁴. The project includes embedding inductive coils beneath campus roads and bus stops, enabling electric buses to charge while driving or stopping. With nearly \$20 million in state funding, the university is expanding its fleet to 13 electric buses and installing static and dynamic wireless charging infrastructure to support continuous operation ahead of the 2028 Olympics.

Electra Afikim Depot (Israel)

Electreon's wireless charging system supports up to 32 buses at this commercial depot. The system achieved 99% operational uptime and over 90% energy transfer efficiency, with most buses starting the day with more than 90% charge¹⁵. This deployment demonstrates how daytime wireless charging can reduce peak overnight demand and eliminate the need for standby vehicles.

Sound Transit's Stride BRT Service (Seattle, USA)

InductEV is deploying 48 wirelessly charging electric buses, including 33 double-decker Alexander Dennis Enviro500EVs, for Seattle's new rapid transit service¹⁶. The system uses 13 in-ground 300-kW inductive chargers, allowing buses to recharge at multiple points along their routes. This setup enhances operational flexibility and reduces downtime, making Seattle the first U.S. city to feature double-decker buses with wireless charging.

¹³ <https://electreon.com/>; <https://www.enrx.com/>; <https://hevo.com/>; <https://www.inductev.com/>; <https://witricity.com/>

¹⁴ <https://transportation.ucla.edu/news/ucla-receives-nearly-20-million-electric-transit-projects>

¹⁵ <https://electreon.com/projects/rosh-haayin-electra-afikim-buses>

¹⁶ <https://www.inductev.com/press-releases/sound-transit-to-go-wireless-in-seattle>

InductEV's Broader U.S. Deployments

InductEV's¹⁷ high-power wireless charging technology is also in use across fleets in Indianapolis, Oregon, Martha's Vineyard, and at port terminals like AP Moeller Maersk in New Jersey. These deployments support a range of heavy-duty vehicles and demonstrate the application of the technology in high-power settings.

4.2. Market share distribution from technical perspective

In traditional conductive DC charging systems, the charger manufacturer is responsible for the entire power conversion chain up to vehicle. In contrast, wireless inductive charging systems require a split architecture, redistributing responsibilities between the charger manufacturer and the OEM. The charger manufacturer supplies the ground-based transmitter unit, including the AC-DC conversion, DC-hfAC inverter, and the transmitting coil. The vehicle OEM is responsible for integrating the receiver coil, resonant compensation network, and rectifier. This architectural shift implies that, as wireless charging becomes more widely adopted, charger manufacturers may see a reduction in their share of the overall charging system market. With OEMs taking on a larger portion of the system integration, particularly on the vehicle side, the value chain becomes more distributed.

However, this split in responsibilities is only feasible once industry standards, such as SAE J2954 and ISO 15118, have reached sufficient maturity and adoption. In the absence of fully established and interoperable standards, OEMs are likely to rely on charger manufacturers to deliver a complete, validated wireless charging system, including both infrastructure and vehicle-side components.

As standards evolve and become widely accepted, the industry is expected to shift toward a more modular and interoperable model, enabling OEMs to take greater ownership of vehicle-side integration while relying on standardized interfaces and protocols for seamless operation across different charger platforms.

4.3. Technology Maturity and Diffusion

To assess the maturity and diffusion of wireless charging technology, a Bass Model analysis was conducted¹⁸, integrating indicators from academic research, commercial activity, and intellectual property trends. The findings reveal a complex landscape where technological development has outpaced market adoption, and where public interest is only now beginning to align with earlier innovation efforts. Figure 21 illustrates the

¹⁷ <https://www.inductev.com/>

¹⁸ Takahashi, C. K., de Figueiredo, J. C. B., & Scornavacca, E. (2024). Investigating the diffusion of innovation: A comprehensive study of successive diffusion processes through analysis of search trends, patent records, and academic publications. *Technological Forecasting and Social Change*, 198, 122991.

diffusion levels according to the number of academic publications¹⁹, patent applications and publications²⁰, and search interest scores of the general public in search engines²¹.

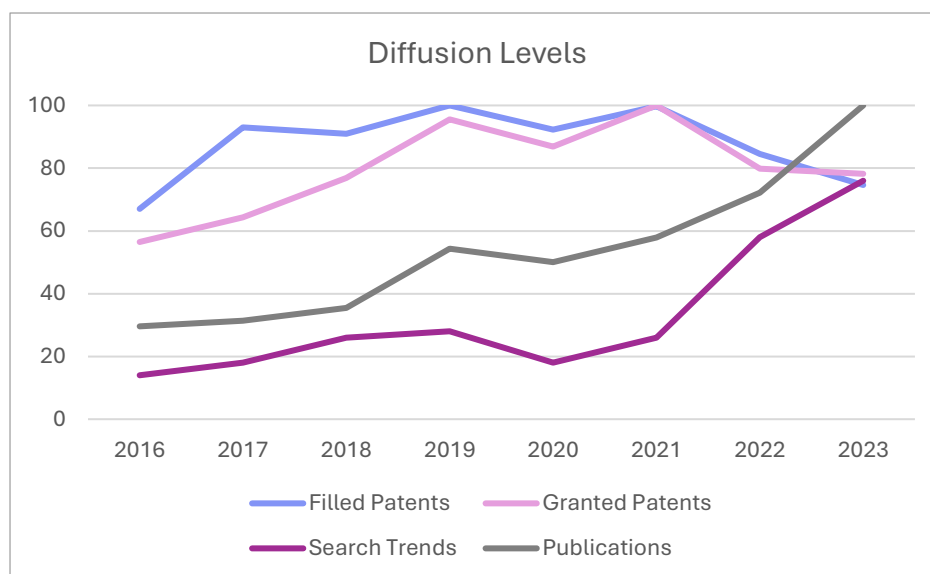


Figure 21. EV Wireless Charging Maturity and Diffusion.

Key Observations

- There has been consistent growth in academic and consumer interest, as evidenced by rising scientific publication counts and internet search trend scores.
- Patent applications surged around 2010, marking a period of intense technological development. However, market adoption and academic engagement lagged by nearly a decade, indicating a gap between innovation and commercialization.
- Since 2021, patent applications have slightly declined, while the number of granted patents has stabilized, suggesting a plateau in new technological breakthroughs.
- Technologies such as coils and control systems have reached a level of patent maturity, while areas like dynamic charging, foreign object detection (FOD), and interface systems remain in active development, presenting growth opportunities.

Interpretation of Trends

The recent increase in public interest suggests that the market may be entering a phase of early adoption, following years of foundational development by manufacturers and technology providers. However, the slight decline in patent activity since 2021 raises questions about the industry's long-term commitment to advancing wireless charging technologies.

Several hypotheses may explain the observed decline in patent activity and industry momentum:

¹⁹ <https://app.dimensions.ai/>; <https://scholar.google.com/>; <https://www.webofscience.com/>

²⁰ <https://worldwide.espacenet.com/>; <https://patents.google.com/>

²¹ <https://trends.google.com/trends/>; <https://www.bing.com/webmaster/>

Market Readiness Gap: The technology may have matured before the market was ready to adopt it, resulting in delayed commercialization and reduced industry enthusiasm.

Technical Feasibility Challenges: Some patented technologies may have proven technically unfeasible or impractical to scale by the time they were ready for deployment.

Integration and Standardization Barriers: Wireless charging may have faced challenges in standardization, vehicle integration, and cost reduction, making it difficult to generalize across platforms and use cases.

Safety Concerns: The transmission of high power across an air gap raises safety concerns, particularly regarding electromagnetic exposure and foreign object risks. These concerns may have slowed consumer and business adoption until robust safety solutions are in place.

4.4. Market Players

This report categorizes the EV wireless charging market players into four roles, i.e., Pure Players, Traditional OEMs, Technology and Component Suppliers and Infrastructure and IP Providers, which were described in previous chapters. Table 6 describes in detail the role of each player and provides examples of the leading organizations on each category. In addition, Appendices 2 to 6²² describe technologies, products, achievements and metrics, and pilot projects of relevant Pure Players.

Table 6. EV wireless charging market key players by role

Market Role	Description	Companies
Pure Players	Their core business is dedicated to developing, deploying, or licensing innovative charging solutions Their offer ranges from stationary pads to dynamic road-embedded systems.	• WiTricity • Electreon • InductEV • ENRX • Elonroad
Traditional OEMs	Established vehicle manufacturers where wireless charging is part of their interests. Often engaged in research or pilot projects, it is not yet a strategic pillar in their broader electrification roadmap.	• Toyota • Hyundai • Honda • BMW • Volkswagen
Technology and Component Suppliers	Offer automotive-grade components or integrated systems, including wireless charging modules. Wireless charging is only a subset of their wider product portfolio, still their role is critical for mass-market adoption.	• IHI • Mahle • Bosch • Hitachi • Siemens

²² <https://electreon.com/>; <https://www.enrx.com/>; <https://hevo.com/>; <https://www.inductev.com/>; <https://witricity.com/>

Infrastructure and IP Providers	Firms that offer enabling technologies, intellectual property, or infrastructure-level systems that support wireless charging. Serve as foundational players through licensing, system integration, or grid solutions.	• Panasonic • LG Group • Samsung • KAIST • Guangxi Power Grid
---------------------------------	--	---

Geographical spread of the players

According to their role, some players have been leading the market and IP development in particular areas. For instance, American pure players have dominated the practical application of wireless technologies. However, Chinese and Korean companies have been leading in terms of patents. Below, Figure 22 represents the regional key players by company roles in the wireless charging market is illustrated.

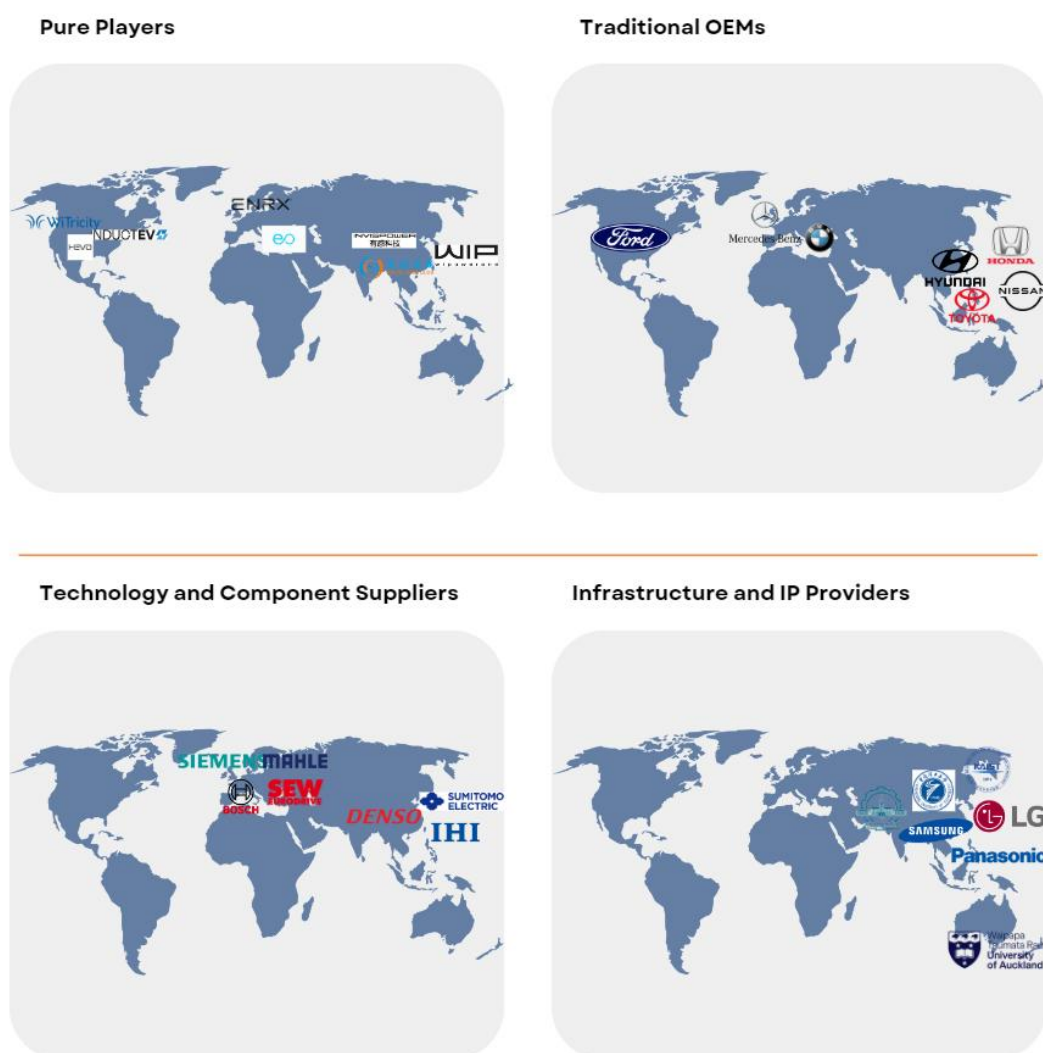


Figure 22. Regional key players by company roles in the EV wireless charging market.

Pure Players

In Pure Players, there is considerable market saturation and maturity in low-power, static charging, followed by static high-power, whereas dynamic charging has been in the development phase with some commercial applications, particularly for low-power applications. High-power dynamic charging, a less mature technology, is currently in development phases, with potential for further applications. Figure 23²³ shows the application focus of each player according to power level and charging type, i.e., static and dynamic.

Highlights

- ENRX and Electreon agreement to advance high-power SAE J2954 standard.
- ENRX and Electreon offer the highest power levels, 200 and 100 kW, respectively.
- Wireless efficiencies between 90-95%, while conductive transfer reaches > 95%.
- Growing public-private investments in pilot projects, ranging from \$2 to \$14 million.
- Recent patents on adaptive power, thermal management, and dynamic charging.
- Witricity and HEVO alignment tolerance in low-power, with potential for high-power.
- Growing Chinese EV industry, Invispower and Anjie Wireless becoming relevant.
- Advances in high-power maritime, agriculture, and material handling industries.
- Vehicle-to-Grid (V2G) considered for low-power applications (Witricity and HEVO).

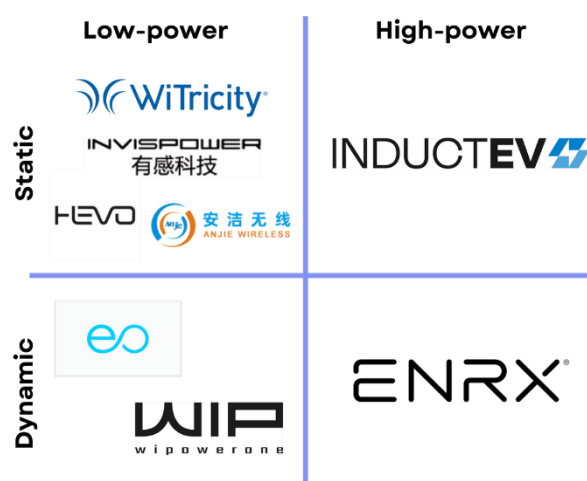


Figure 23. Application focus of Pure Players according to power level and charging type.

²³ <http://www.anjiecorp.com/>; <https://electreon.com/>; <https://www.enrx.com/>; <https://hevo.com/>; <https://www.inductev.com/>; <https://www.invispower.com/>; <http://wipowerone.com/>; <https://witricity.com/>

5. Summary

Despite the promising potential of wireless charging for electric vehicles (EVs), several critical challenges hinder widespread adoption, especially at the industrial scale. These challenges are categorized into three domains: standardization and regulatory gaps, technical challenges, and market and deployment barriers.

5.1. Standardization and Regulatory Gaps

Wireless charging standards have reached a mature stage primarily for static, low-power applications. However, there remains a notable absence of harmonized specifications and clear installation regulations for high-power and dynamic use cases.

- Current standards are only mature for static low-power (<50 kW) applications.
- Most standards focus on static, low-power applications; dynamic and high-power specs are immature.
- Dynamic charging and high-power use cases still lack harmonized global standards as not all standards are yet ready. Many standards are still undergoing development such as SAE J2954/2 and IEC PAS 61980-4 for high-power, and J2954/3, even though IEC PAS 61980-5 for dynamic was already released 2024. Also, ISO 5474-4:2025 defines vehicle side if the infrastructure is available.
- Infrastructure installation regulations, especially for road-embedded systems, remain fragmented and unclear.
- Fragmentation between regional standards delays global interoperability.
- Lack of universal mechanical interface specifications between vehicles and ground systems.

5.2. Technical Challenges

Wireless charging technology encounters several engineering challenges that significantly influence its development. These challenges encompass inefficiencies in energy transfer, constraints related to thermal management and Electromagnetic Compatibility (EMC), and concerns regarding the durability and safety of the system, particularly in dynamic and high-power applications.

- Efficiency drops due to coil misalignment and air gaps, especially in dynamic scenarios.
- Misalignment between transmitter and receiver coils can drastically reduce power throughput.
- Thermal management in charger pads is complex due to space limitations and material temperature range limitations.
- Electromagnetic Compatibility (EMC) and field leakage raise safety concerns, particularly with larger air gaps.

- EMC regulation compliance is especially challenging at high frequencies and power levels.
- Vehicle-side complexity: integration adds weight and requires robust onboard components like converters and shielding.
- High-frequency operations (85 kHz) limit cable length and require premium-grade wiring.
- Durability of infrastructure components (e.g., road-embedded coils) under snowplowing, weather, and heavy-duty usage.
- Foreign Object Detection (FOD) is under development and presents safety limitations.
- Dynamic charging requires precise and synchronized power handover between coils while the vehicle is in motion.

5.3. Market and Deployment Barriers

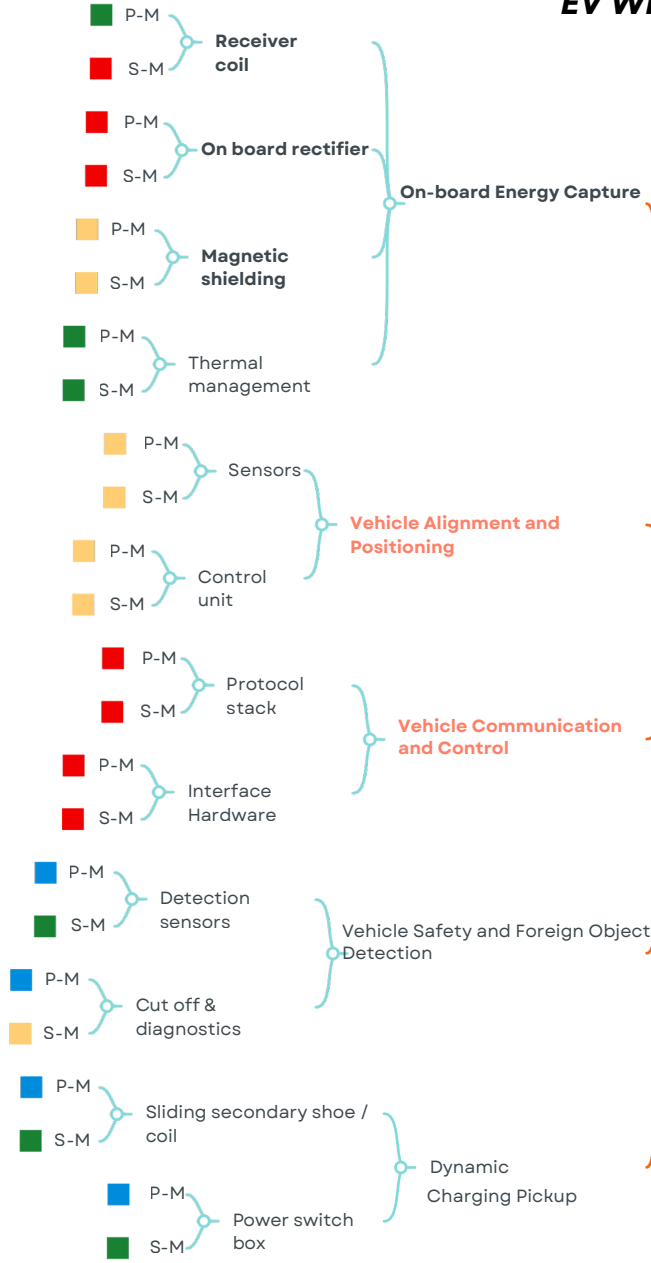
Significant infrastructure costs, limited interoperability, uncertain return on investment, and a lack of public trust obstruct the widespread implementation of wireless charging, particularly in challenging environments.

- Infrastructure costs for dynamic systems and public road modification are substantial.
- Interoperability limitations between systems and vehicles due to a lack of universal interface standards.
- Social trust and perception of safety and effectiveness need to be developed.
- Cost parity with cable-based DC fast charging has not yet been reached for industrial deployments. Due to technical requirements, cost parity will be challenging to reach.
- Without large-scale deployments, the ROI of wireless charging for fleet fleets is difficult to quantify. Despite potential advantages in automation and reduced wear, current efficiency limitations relative to conductive charging may offset these benefits.
- Deployment in heavy-duty segments (ports, mining) faces harsh operating environments.

6. Appendixes:

EV Wireless Charging Technologies and Key Components

Patent and Standardization Maturity



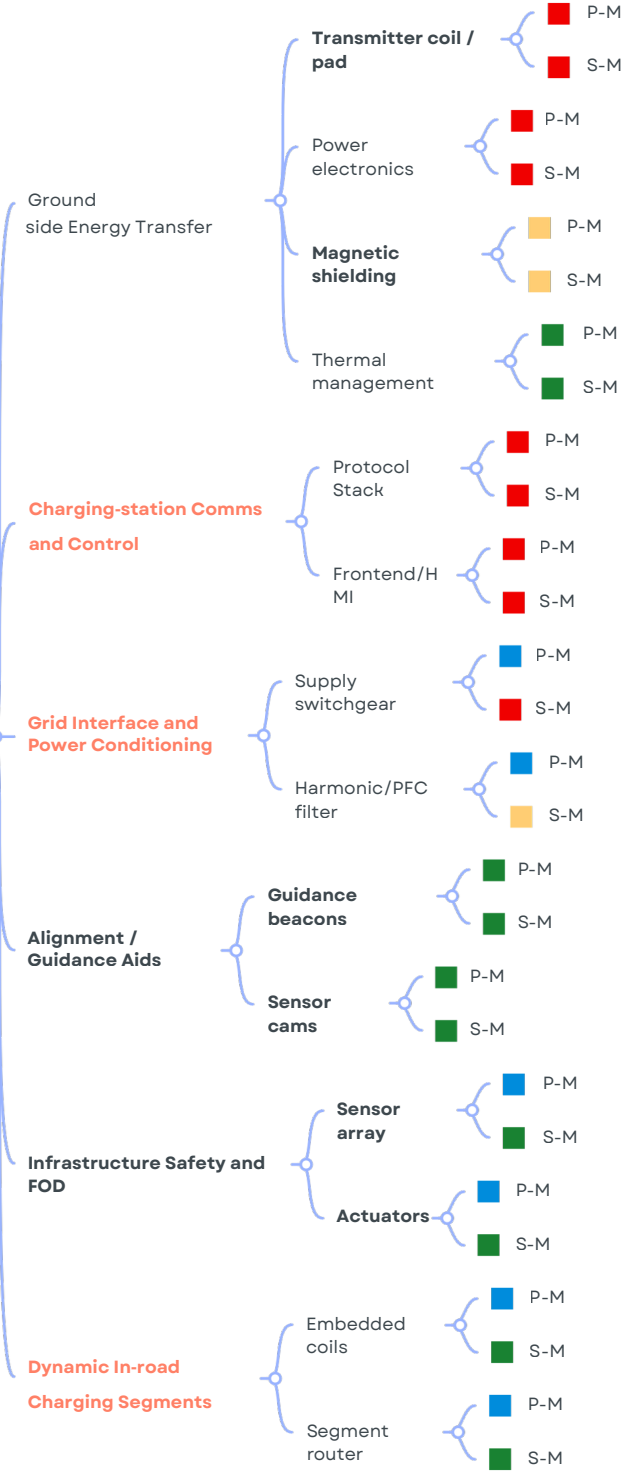
Standard Frequency
86 kHz

Technologies

Maturity Scales

- Patent Maturity (P-M)**
- Saturated (> 50 patents)
 - Mature (26-50 patents)
 - Growth (11-25 patents)
 - Emerging (≤ 10 patents)
- Standardization Maturity (S-M)**
- High (fully-published, widely cited)
 - Medium (draft/partial specs)
 - Low (early WG stage / guidelines only)
- Maturity Scales

Infrastructure-side technologies



Company Overview

Business Model

Technology and IP licensing to:
Automotive OEMs and Tier 1 suppliers.

Performance Metrics

IP that supports **SAE, ISO, and GB**.
Transf. efficiency through materials **90%**.

Standardization Overview

Active role in developing **SAE J2954**.
Licensed tech. as standard reference.

Patent Overview

245 patents - focus transfer and FOD.
Recent patents - signaling and sensing.

Technology Overview

Product Portfolio

WiTricity Halo™ platform

- Ground-based charging pad and a vehicle-mounted receiver.
- It transfers power efficiently through materials like stone, cement, and asphalt.
- Charging rates vary from 3.6 kW to 11 kW for passenger vehicles.
- Platform scalable up to hundreds of kW, making it suitable for high-power.

Magnetic Resonant Coupling

Hardware

Halo™

- 3.6 to 11 kW
- High-power scalable

Power Transfer

- Spatial freedom
- Allow misalignment

Power Flow

- Vehicle-to-Grid
- Bidirectional

Company Highlights

- Pioneer of magnetic resonance-based wireless charging in 2010.
- In 2019, Qualcomm acquired Qualcomm's Halo division, including its IP portfolio.
- Besides patents, it has filed more than 1000 technology designs and licences.
- Customer model is mature and B2B-focused on IP licensing to OEMs and suppliers.

Aspect	Description
Product Maturity	Commercial Product stage (evidenced by factory integration into commercial vehicles)
Pilot Projects	4 high-profile projects in USA and China
Estimated Annual Revenue	\$10 million to \$50 million

Pilot Project - Smart Parking Lab (Detroit, Michigan, USA)

Use case to demonstrate wireless charging in automated parking, as autonomous vehicles will require a method that eliminates the need for manual cable plugging.



Project Timeline

Partnership Announcement
Februray 2022

Public collaboration announcement

Ford and Bosch also as project partners

Project Deployment
Summer 2022

Proof-of-Concept
2023

Halo demonstration for autonomous EV

Ready for commercialization

Project Completed
2023

Company Overview

Business Model

High-power and dynamic charge.
Commercial and Industrial.

Performance Metrics

200 kW in static and dynamic, **90% eff.**
180 kW transfer at **50 km/h.**

Standardization Overview

Active role in **SAE J2954** for high-power.
Member of multiple std. committees.

Patent Overview

13 patents - inductive power tech.
Recent - Induction for water vehicles.

Technology Overview

Product Portfolio

PRIMOVE

- High-power static charging systems for buses and trucks up to 300 kW
- Dynamic Electric Roadway system

ENRMOVE

- Wireless charging system for intralogistics market, such as AGVs, forklifts, and autonomous robots.

PWRMOVE

- Contactless power for rail-guided industrial applications, such as conveyors, sorters, and monorails.

Inductive Resonant
Power Transfer

Hardware

Three-phase coil
Under road embed
Uniform magn. field

Static Charging
Automated charge
Trucks and heavy eq.

Electric Roadway
Dynamic Charging
High-power industrial

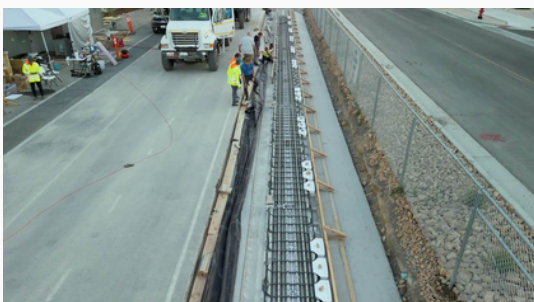
Company Highlights

- Long-standing expert in industrial heating and IPT Technology.
- Owned by a Norwegian investment firm focused on energy and technology.
- Emerging business focused on the intralogistics and industrial automation markets.
- Pilot projects focused on high-power public buses with power ranges 100-300 kW

Aspect	Description
Product Maturity	Commercial Product stage (PRIMOVE, ENRMOVE and PWRMOVE)
Pilot Projects	10 documented projects (Europe mainly)
Annual Revenue	\$178 million USD (Mainly focused on industrial heating)

Pilot Project - Central Florida Expressway (Orlando, USA)

Partnership with CFX and ASPIRE Research Center to deliver the world's first 200 kW 8 km electrified roadway supporting long-haul trucks and passenger vehicles while in motion.



Project Timeline

**Partnership
Announcement**
2022

Heavy and light
powered vehicles

\$14 million, incl. test
and pilot tracks

Initial Investment
2023

Deployment
2023-2025

Initial testing on a
40-meter test track

Beginning of
segment
construction.

**Expressway
construction**
2026

Company Overview

Business Model

High-power hardware and system:
Commercial and industrial fleets.

Performance Metrics

Grid-to-battery efficiency **90%**.
Same as high-power **wired** charger.

Standardization Overview

Developing **SAE J2954/2** high-power.
In process to standarize up to **500 kW**.

Patent Overview

16 patents - *trans. and alignment.*
Recent - thermal mgmt. and dynamic.

Product Portfolio

- Hardware** consists of an In-Ground Charger, an On-Vehicle Receiver, and a Power Cabinet
- It can be placed up to 100 feet away to conserve valuable real estate (75-450 kW).
- AI-powered **software** platform .
- Sensor telematics for monitoring fleets, energy optimization, and predictive maintenance.

Technology Overview

High-power Inductive Charging

Hardware

Ground Assembly
75 kW ground pad
Modular set-ups

Power Cabinets
Up to 100 feet away
Diff. weather cond.

Vehicle Assembly
Receiver pad
Undercarriage

Software

InductAI
Platform design
Fleet mgmt.

Intelligent Control
Charging process
Data logging

Data Transfer
Secure comm. link
High-speed

Company Highlights

- Formerly known as Momentum Dynamics, it focuses on high-power charging solutions.
- Leader in the demanding commercial and industrial EV fleets market.
- It has multiple, large-scale deployments with major customers.
- All its customers are expanding operations with wireless once pilots are done.

Aspect	Description
Product Maturity	Fully Commercial Product stage (Hardware and Software solutions)
Pilot Projects	9 distinct deployments in North America and Europe
Estimated Annual Revenue	\$15 million USD

Pilot Project - Port of Long Beach (Long Beach, California, USA)

Partnership with the Port of Long Beach to install high-power wireless chargers, allowing five battery-electric-cargo-handling vehicles to charge automatically while parked.



Project Timeline

Partnership Announcement
2024

Five up e-cargo vehicles up to 75 kW

Proof-of-Concept
2025

Wireless inductive char

Two charging stations and 20 EVs

Project Deployment
2025

\$ 3.3 million grant by California govnt.

Live Project
2025

Company Overview

Business Model

Dynamic Charging as a Service:
Recurring fee to fleet operators.

Performance Metrics

Car receivers for **7, 11, and 25 kW**.
Up to **100 kW** transfer at **80 km/h**.

Standardization Overview

Developing **SAE J2954/3** dynamic chrg.
Diff. from standard **J2954 guidelines**.

Patent Overview

10 patents - dynamic charging.
Recent - current regulation and ctrl.

Technology Overview

Product Portfolio

- Full-stack solution that includes three core components:
 - Underground infrastructure (in-road copper coils).
 - Vehicle-side technology (receivers installed on EVs).
 - Cloud-based management software platform for live monitoring and smart charging insights.
- Applications for dynamic charging, semi-dynamic charging, and static charging

Adaptive Inductive Charging

Hardware

Underground Coils

- Road installed
- Below asphalt surf.

Mngmt. Units

- Roadside cabinets
- Mngmt. power flow

Vehicle Receivers

- EV undercarriage
- Capture power

Company Highlights

- Dominant force in wireless charging for public infrastructure and commercial fleets.
- Provides a comprehensive, end-to-end charging solution
- It has pioneered the use of dynamic wireless charging, known as an Electric Road System (ERS).
- Its customer base is highly mature, consisting primarily of government bodies, public transit, and logistics operators.

Aspect	Description
Product Maturity	Commercial Product stage (numerous live, operational projects globally,)
Pilot Projects	At least 10 major international projects
Estimated Annual Revenue	\$31,65 million USD Growth 37%

Pilot Project - Michigan Department of Transportation (Detroit, USA)

Partnership with the state government agency and Ford Motor Co., representing the first public wireless charging road for heavy (up to 25 kW) and light (7-11 kW) vehicles.



Project Timeline

Partnership Announcement
2022

Multiple Ford E-Transit test vans

Up to \$1.9 million + additional funding

Initial Investment
2023

Deployment
2023

Quarter mile segment (66 coils)

Data gathering and test performance

Live Testbed Road
2025

Company Overview

Business Model

Hardware and Systems Sales to:
Commercial and Automotive OEMs.

Performance Metrics

Grid-to-battery efficiency **95%**.

Outlet-to-battery efficiency **91.5%**.

Standardization Overview

Compliant **UL 2750** and **SAE J2954**.

Member of SAE J2954 committee.

Patent Overview

5 patents - focus FOD and alignment.

Recent patent on **dynamic** charging.

Technology Overview

Product Portfolio

- **REZONANT E8**, a Level 2 charger that delivers up to 8 kW of power
- **"Guardian"** living and foreign object detection system, which uses specialized infrared technology.
- AI-powered **software** platform .
- **JOURNEY** app to handle station reservation, vehicle alignment, remote charging control, and payments.

Magnetic Resonant Coupling

Hardware

Charging Power

- Up to 8 kW Level 2
- AC/DC Compatible

Power Pad

- Surface mounted
- GAWR 20,000 lbs

Vehicle Assembly

- 8 kg receiver
- A/M installation

Software

Mobile App

- Station booking
- Alignment guidance

Remote Functions

- Remote monitoring
- Session control

Asset Mgmt.

- Fleet monitoring
- Station mgmt.

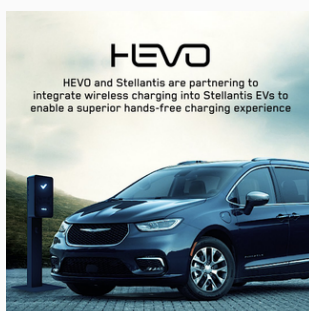
Company Highlights

- Focus on developing a cost-effective and accessible wireless charging solution for urban environments.
- It targets a broad customer base, including fleet operators, automakers, and individual consumers for home and workplace charging.
- Its go-to-market strategy relies heavily on forging strategic partnerships to integrate and deploy its technology, and thus become a Tier 1 supplier.

Aspect	Description
Product Maturity	Commercial Pilot / Early Commercial (pending certification)
Pilot Projects	At least 5 major international projects
Estimated Annual Revenue	Financial metrics unavailable

Pilot Project - Stellantis Partnership (Auburn Hills, Michigan, USA)

OEM integration by offering factory-installed options across Stellantis's brands, i.e., Jeep, Fiat, Maserati, Peugeot, to transform HEVO from a technology provider into a Tier 1 supplier.



Project Timeline

